Power System Protection

How relays detect faults, how circuit breakers respond, and how protection schemes stay selective — from primary defence to intelligent IEDs.

The Stakes: Faults Must Be Cleared Fast

Every millisecond a fault persists, it risks injury, equipment destruction, and cascading outages across the network. Protection systems exist to detect abnormalities and isolate the **smallest possible portion** of the system — quickly, reliably, and without hesitation.

What Protection Really Guarantees

Relays monitor system parameters continuously — voltage, current, power, and frequency — and must act dependably when genuine faults occur, whilst remaining stable during normal disturbances.

Selectivity

Isolate only the faulted zone

Stability

Stay inert to external faults

Speed

Minimise fault duration

Sensitivity

Detect even minimum fault current

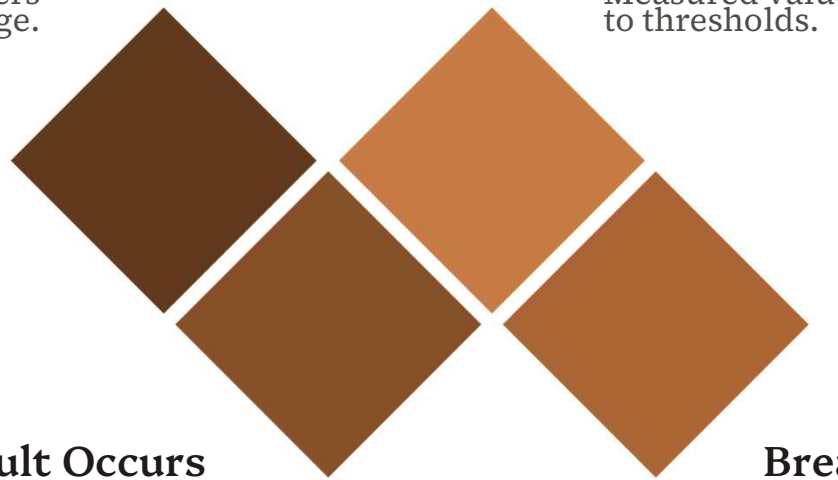
The Fault Journey: What Happens in Real Time

CT/PT Sense

Instrument transformers capture the change.

Relay Decides

Measured values compared to thresholds.



Fault Occurs

Voltages collapse and currents surge.

Breaker Trips

Circuit breaker interrupts current flow.

The entire sequence — from fault inception to current interruption — typically unfolds within **3–10 cycles** on a modern system.



Who Does What: Relay vs Breaker

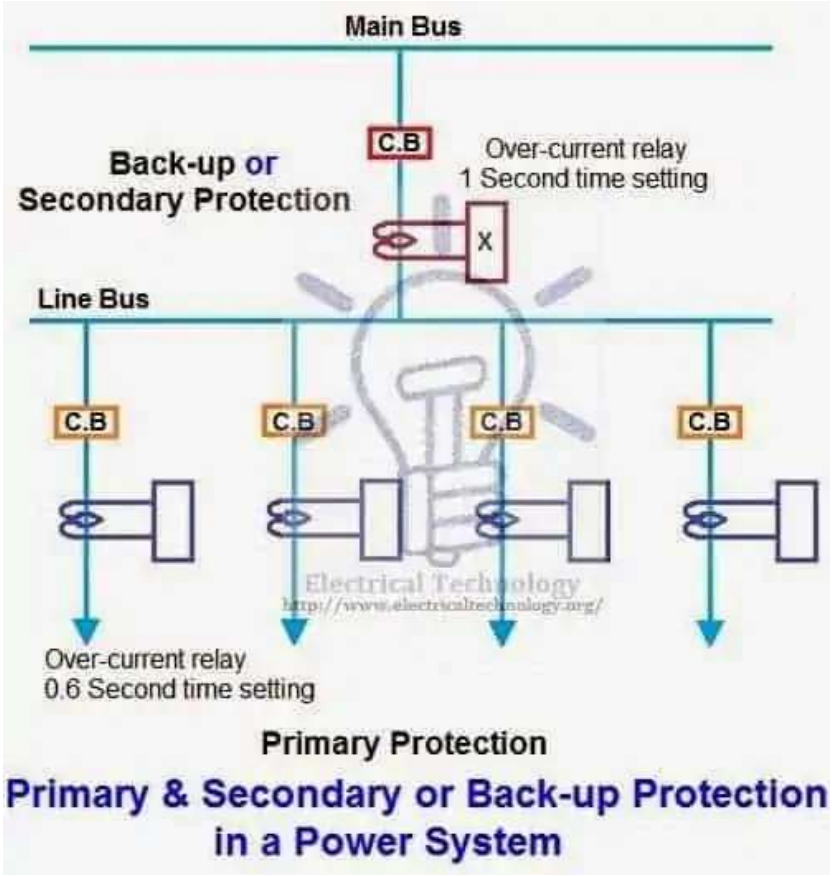
The Relay

Continuously monitors, detects the fault condition, and issues a **trip command** to the breaker coil. It is the brain — measuring, comparing, and deciding.

The Circuit Breaker

Physically interrupts the fault current. It is the muscle. **Total fault clearing time = relay operating time + breaker interrupting time**, expressed in cycles.

Primary vs Back-up: Two Lines of Defence



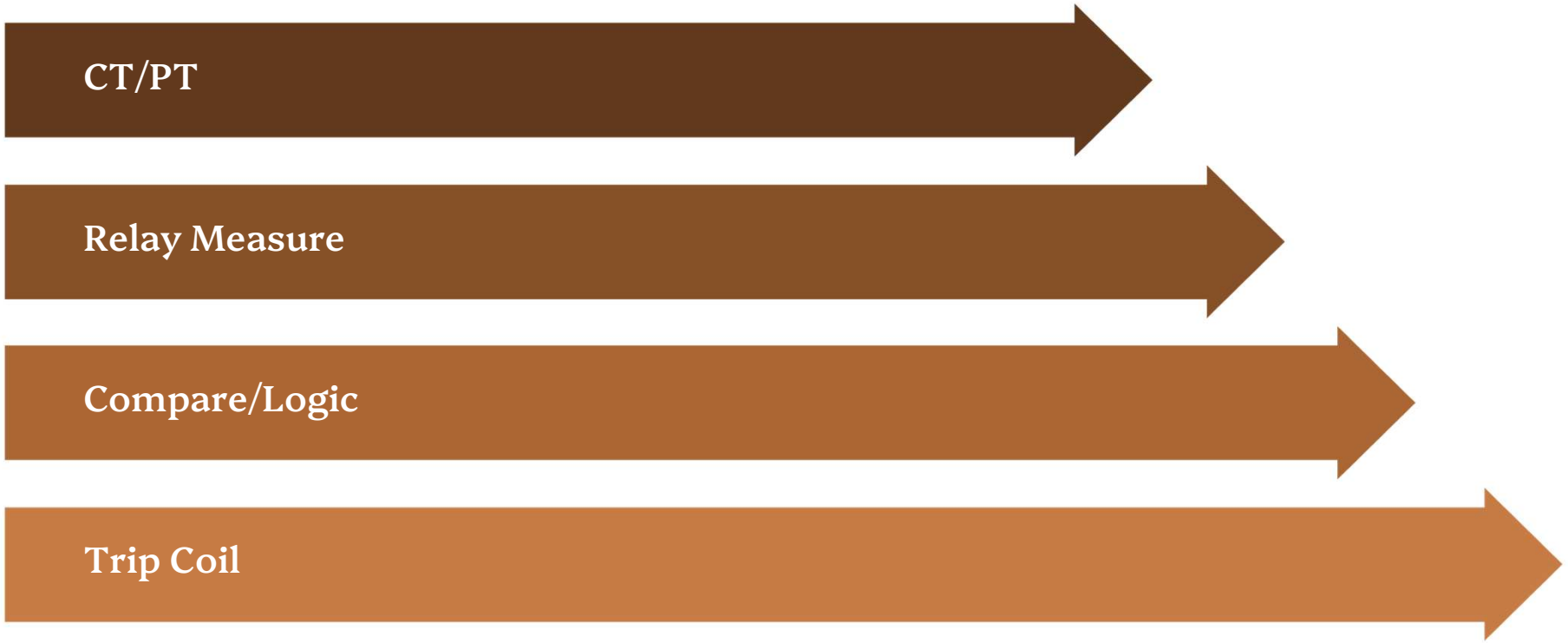
Primary Protection

The first responder. Clears faults within its designated zone as fast as possible — typically instantaneous or with minimal delay.

Back-up Protection

Operates only if the primary scheme or its circuit breaker fails. Accepts a deliberate time delay to give primary a chance to act first. May trip additional breakers to achieve isolation.

Inputs, Outputs, and the Control Wiring Chain



Relay output contacts energise the breaker trip coil via the DC control supply. Separate alarm contacts can alert operators without tripping the circuit.

The Core Protection Requirements: The 4 'S'



Selectivity

Only the breaker **nearest the fault** should trip. Achieved through current-, time-, or zone-based discrimination between adjacent relays.



Stability

Must remain inert during heavy load swings and external faults outside its protection zone — no unnecessary tripping allowed.



Speed

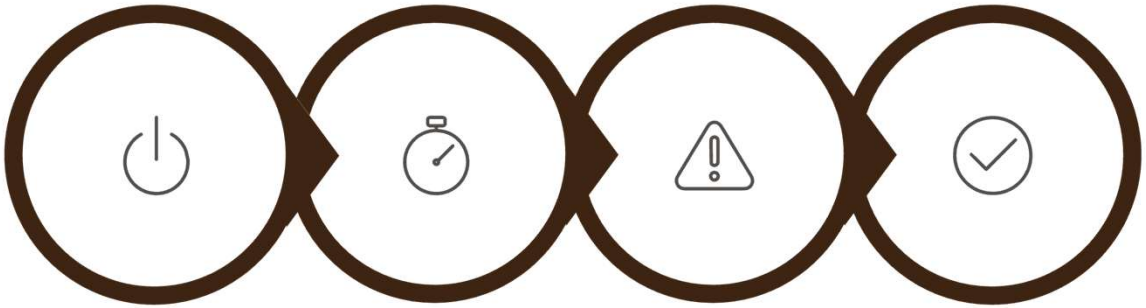
Rapid fault removal limits damage and prevents voltage collapse or loss of synchronism across the wider network.



Sensitivity

The minimum operating current must be set low enough to detect the **smallest credible fault**, even at the far end of a zone.

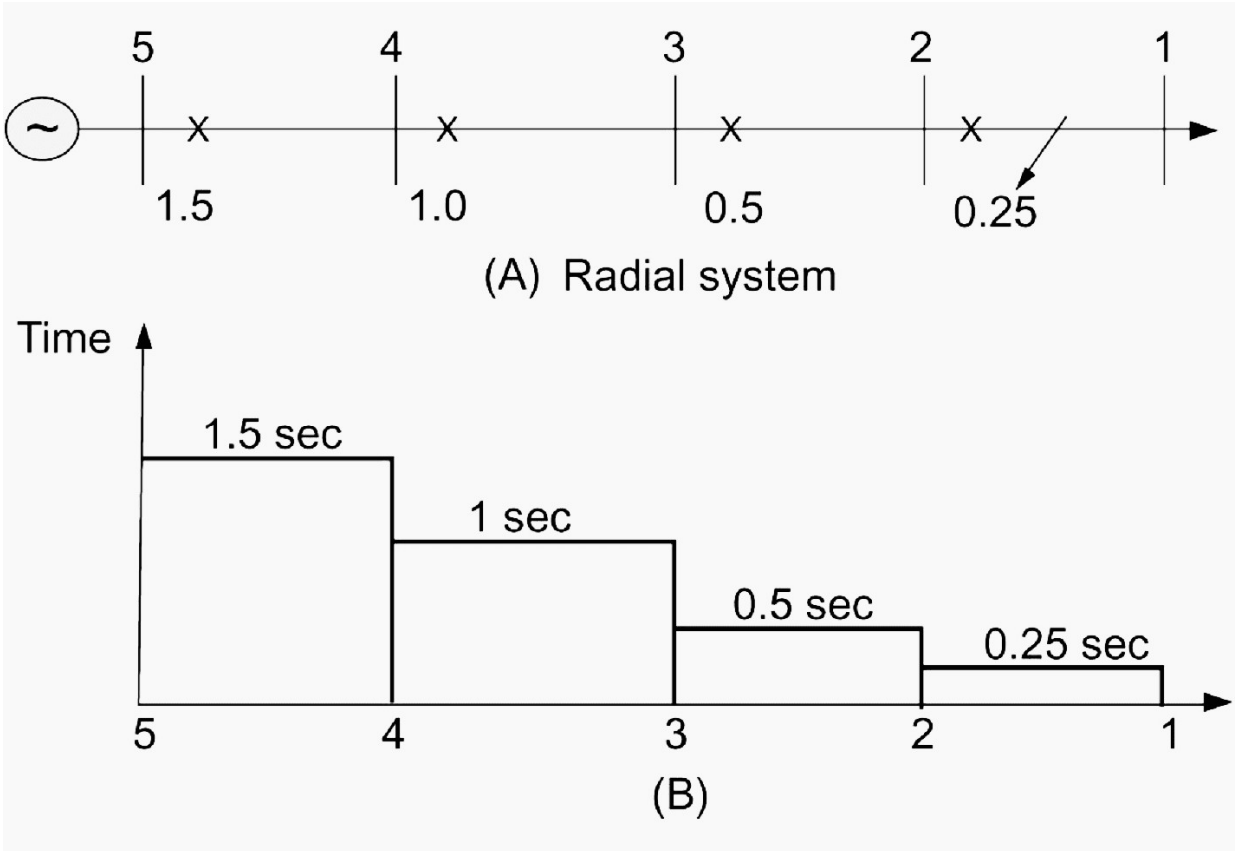
Breaker Failure: The "If It Didn't Open..." Mechanism



Why 15–17 Cycles?

This window allows a healthy breaker adequate time to fully interrupt arc current. Only if current **persists beyond this threshold** does the breaker failure scheme escalate to wider tripping. Once current drops below the preset check level, the initiation is automatically cancelled to avoid unnecessary cascading.

Relay Co-ordination - Time Grading



How Grading Works

Adjacent relays are assigned **staggered time delays** so the relay closest to the fault always operates first. Each upstream relay waits a coordination time interval — typically **0.2 to 0.5 seconds** — before it will operate as back-up.

Why Inverse-Time Matters

Relays with inverse-time characteristics automatically trip faster for higher fault currents, providing both speed at the source and a natural grading margin further upstream.

Relay Co-ordination - Current Grading:

Selectivity by Fault Magnitude

Setting the Pickup

Each overcurrent relay is set with a **pickup current** above maximum load but below the minimum expected fault current. This ensures normal operation never causes a trip, whilst genuine faults always exceed the threshold

Combined Grading

When current grading alone cannot achieve discrimination — because fault currents at adjacent locations are too similar — **inverse-time characteristics** are added so upstream relays operate later, providing reliable back-up without nuisance tripping.

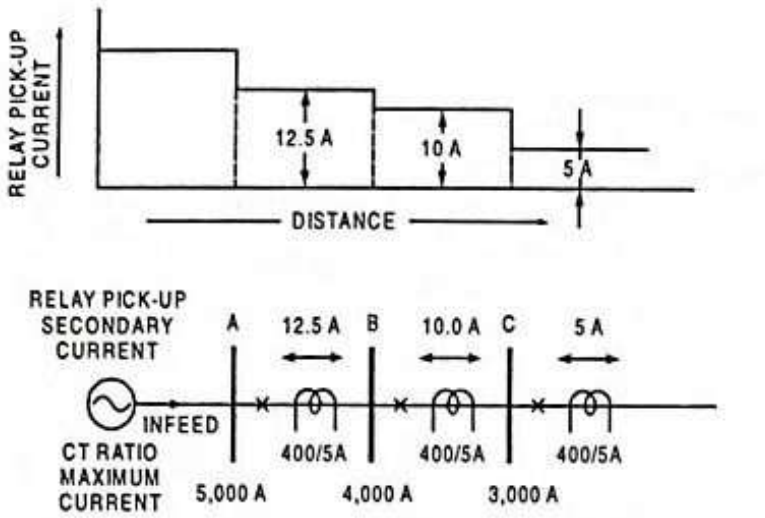


Fig. 14.9. *Current Graded Protection Applied To a Radial Feeder*

The Intelligence Revolution

1 — **Electromechanical Era**

Induction disc and attracted armature relays — mechanical movement to measure current and time. Reliable but inflexible and single-function.

2 — **Static Relay Era**

Solid-state electronics replaced moving parts, improving speed and consistency. Still largely single-function.

3 — **Numerical IED Era**

Microprocessor-based Intelligent Electronic Devices perform multi-function protection, monitoring, metering, and event recording in a single unit with programmable logic (e.g. SELOGIC).



IED (Intelligent Electronic Device) advanced functions that make our life better

The IED brings a relay panel with many single-function electromechanical relays, control switches, extensive wiring, and much more into a single box.

In addition the IED handles additional features like self and external circuit monitoring, real-time synchronization of the event monitoring, local and substation data access, programmable logic controller functionality, and an entire range of software tools for commissioning, testing,

Protection Types in Practice

Magnitude

Overcurrent and undervoltage relays — trip when a quantity exceeds or falls below a threshold.

Directional

Add phase angle discrimination — only operate for fault current flowing in a defined direction.

Differential / Ratio

Compare currents at two or more points — operate when the difference exceeds a fraction of the through-current (biased differential).

Distance (Impedance)

Estimate apparent impedance from V/I measurement to locate the fault within a defined reach zone.

Primary Protection: Different Equipment, Different Principles

⚡ Transformer

Differential protection is the primary scheme — comparing currents entering and leaving the transformer windings. Backed up by Buchholz pressure relay and overcurrent elements.

🔄 Generator

Differential protection again forms the primary scheme, supplemented by loss-of-excitation, reverse power, and stator earth fault protection elements.

ƀ Transmission Line

Distance (impedance) relays and directional overcurrent relays serve as primary protection, with reach zones defined as a percentage of line impedance.

The Foundation: Overcurrent (OC)

Overcurrent protection is the **earliest and simplest** protection method. A relay operates whenever the measured current exceeds a pre-set pickup value. It relies on current magnitude alone, with no directional or impedance awareness — making it perfectly suited to straightforward **radial distribution systems** where fault current flows in only one direction.

Principle

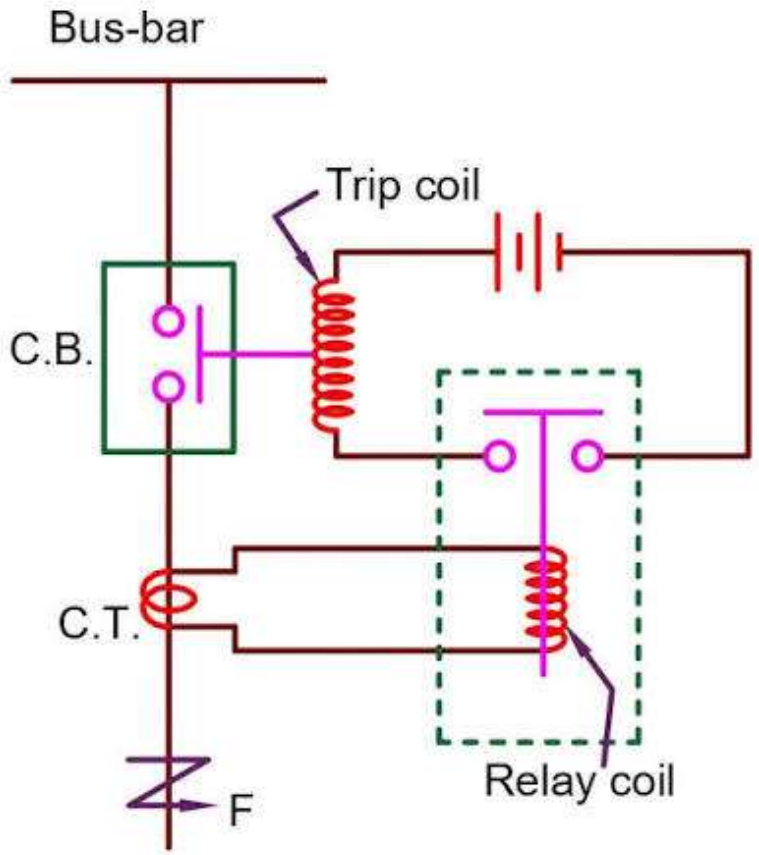
Trip when I > Ipickup

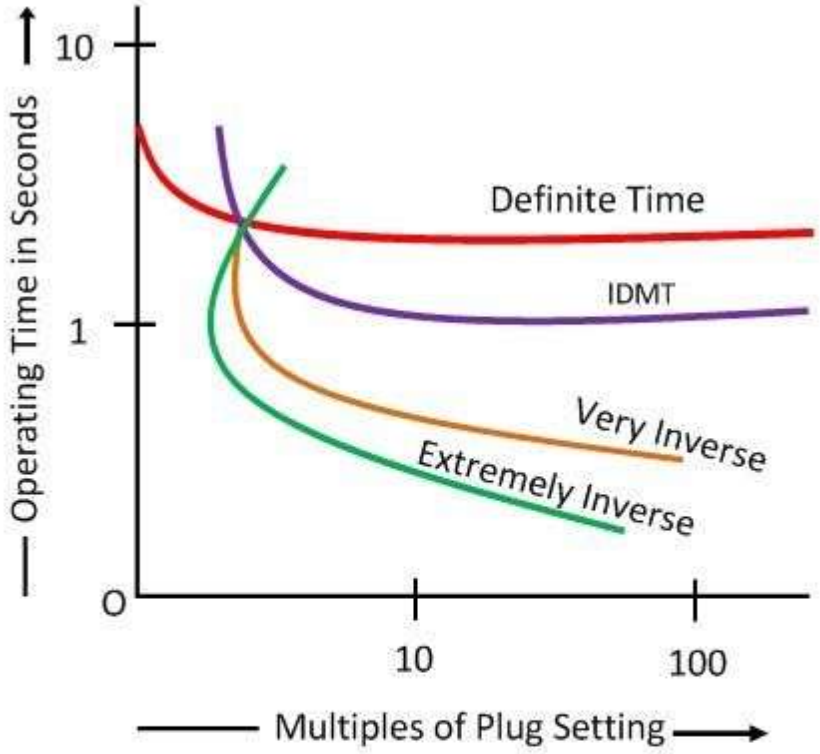
Input

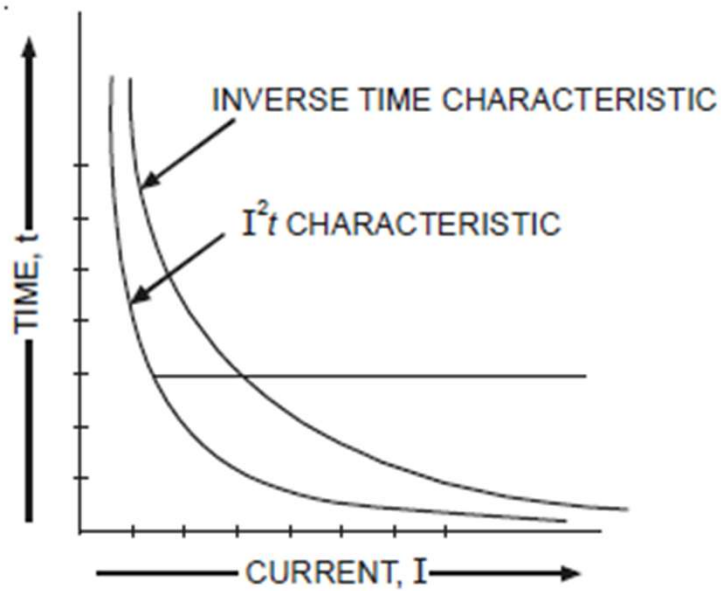
Current magnitude only

Best for

Radial systems with single source

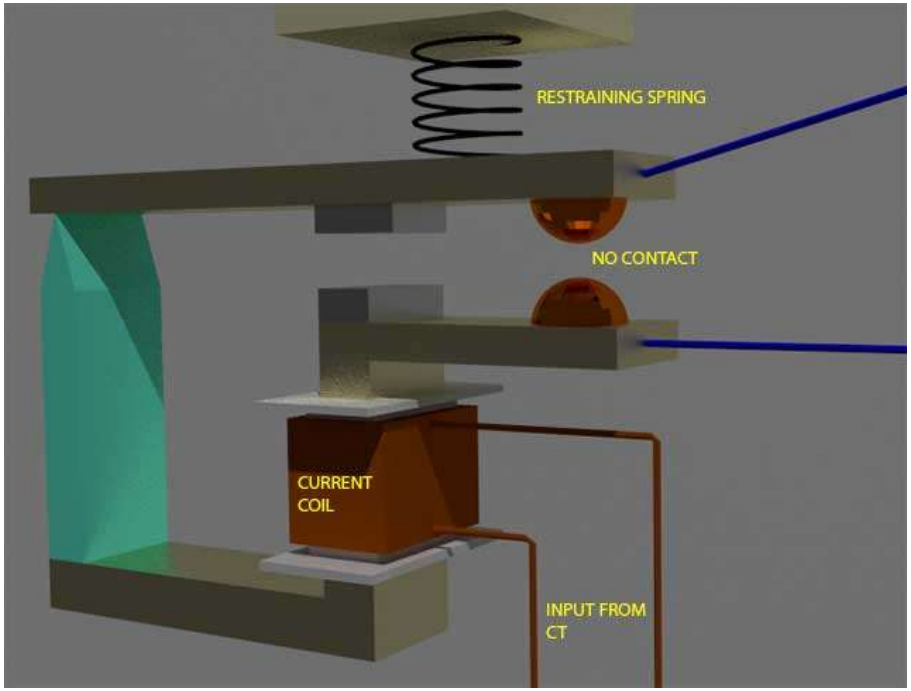

Bus-bar
Trip coil
C.B.
C.T.
F
Relay coil





Characteristic of Various Overcurrent Relay

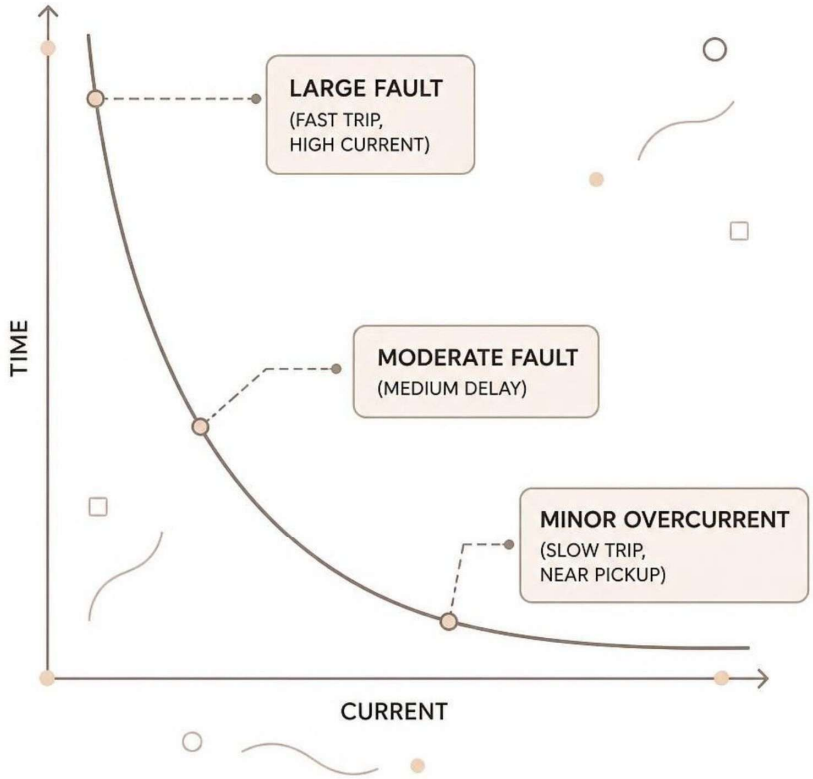
Circuit Globe



Current Coil: Connected to the secondary of a current transformer (CT) on the power line. It produces a magnetic flux proportional to the load or fault current.

Voltage Coil: Connected to the secondary of a potential or voltage transformer (PT/VT). It produces a magnetic flux proportional to the system voltage

Time-Current Characteristics



Inverse Time Relays

Relays trip **faster for large faults** and slower for minor overcurrents — the trip time is inversely proportional to the magnitude of the fault current. This characteristic is defined by standard curves (IEC/ANSI).

Coordination Margins

Adjacent relays are graded with a coordination time interval of **0.2–0.5 seconds**, ensuring only the relay closest to the fault operates, whilst upstream relays act as backup.

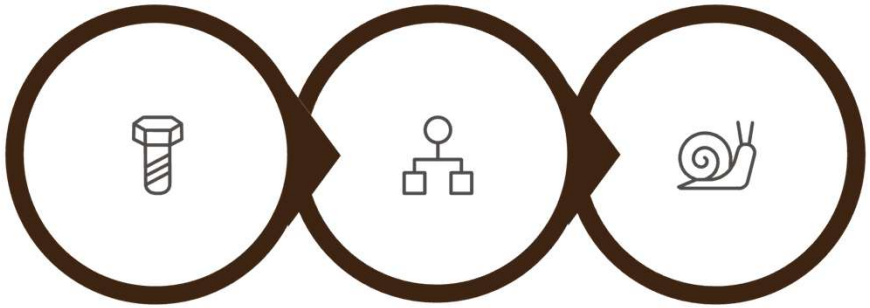
The Radial Constraint

Why Distance Matters

In radial systems, fault current is inversely proportional to the impedance between source and fault. A fault close to the source produces very high current; a fault at the end of a long feeder produces much lower current — and therefore a much slower relay response.

The Layering Problem

As more protection layers (zones) are added, the upstream relay's trip time grows significantly. At the source relay, **excessive clearing times** can result, exposing equipment to prolonged fault energy.



Near Fault

Mid-line Fault

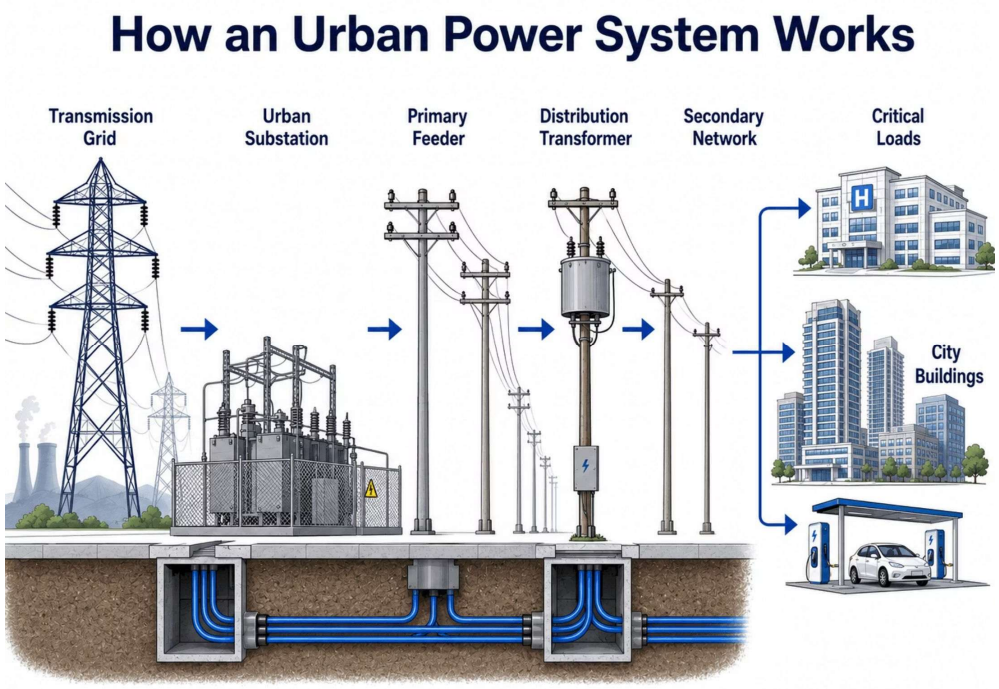
Remote Fault



Directional Protection

When power can flow in multiple directions, current magnitude alone is no longer sufficient to identify a fault's location.

Adding Directionality



Why Direction Is Needed

In **networked or meshed systems**, fault current can flow from multiple directions. A simple overcurrent relay cannot distinguish between a fault in front of it and one behind it — potentially causing unwanted or missed trips.

The Solution

Directional relays assess both **current magnitude and the phase angle** relative to a reference voltage. Only faults occurring within the designated forward direction trigger a trip, blocking operation for reverse-direction faults.

How Directional Relays Work

1 Measure Phase Angle

The relay compares the phase angle between the measured current (I) and a polarising reference voltage (V). The angle determines the direction of power or fault current flow.

2 Define the Trip Region

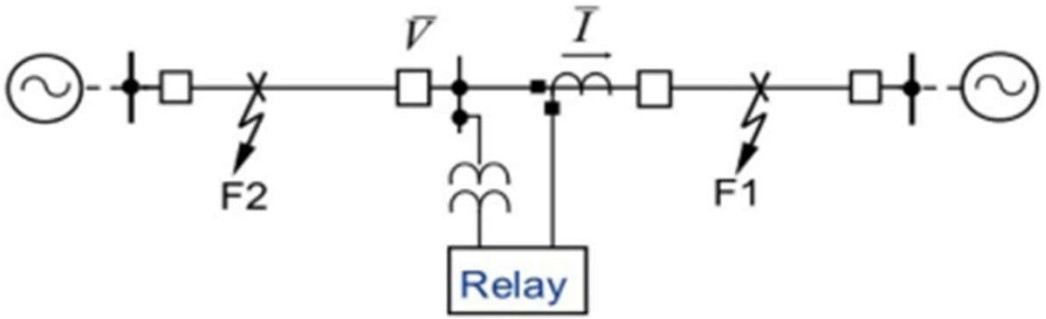
A characteristic angle and a trip boundary define the "forward" operating zone. Faults within this region produce a phase angle that falls inside the trip boundary.

3 Block Reverse Faults

If the measured angle falls outside the forward zone — indicating a reverse fault — the relay is blocked, preventing unnecessary circuit breaker operation.

Directional Overcurrent Protection

Basic Principle



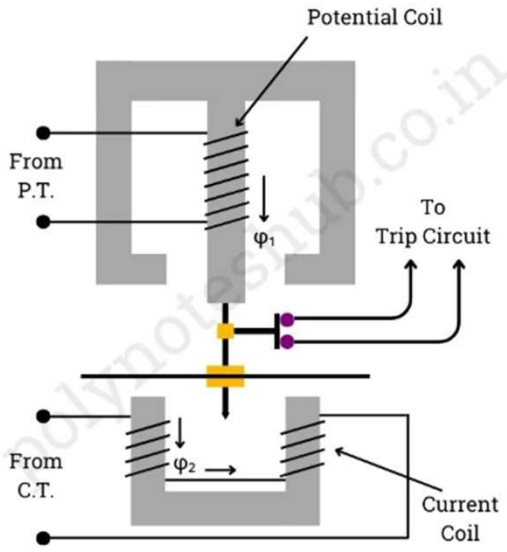
Reverse Fault (F2)



Forward Fault (F1)



Construction of Directional Relay



Construction of Directional Relay

Induction Disc: In traditional electromechanical directional and directional overcurrent relays, an aluminium disc experiences electromagnetic torque produced by the interaction of voltage and current coils (fluxes)

Direction Control: When power or current flows in the normal direction, the torque keeps the disc still or moves it away from the contacts. When the power flow reverses during a fault, the torque reverses and rotates the disc to close the trip circuit

Distance Protection

By measuring the ratio of voltage to current, distance relays can pinpoint where on a line a fault has occurred — independent of system loading conditions.

The Impedance Shift

From Current to Impedance

Distance relays replace simple current measurement with the **ratio of voltage to current (Z = V/I)**. Since line impedance is proportional to length, this ratio directly indicates the electrical distance to the fault — regardless of source impedance or system loading.

Key Advantage

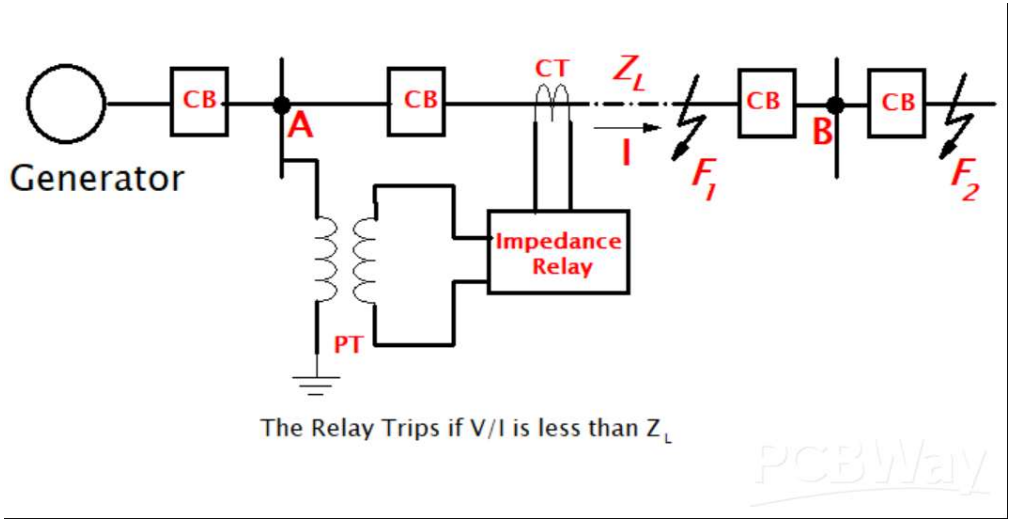
Unlike overcurrent relays, distance relay performance is **largely independent of fault current level**, making it suitable for heavily meshed transmission networks where source impedance varies with system conditions.

The Relay Equation

$$Z_{measured} = \frac{V_{relay}}{I_{relay}}$$

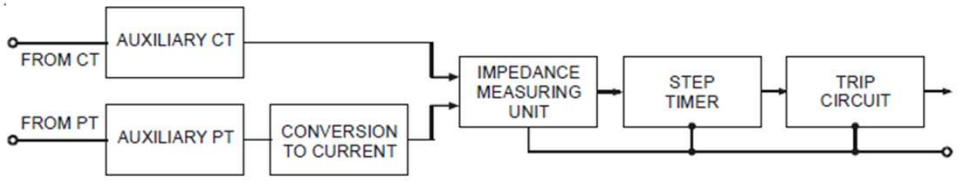
When Zmeasured falls within the relay's reach setting, a fault is detected at that distance on the protected line.

A distance relay works by measuring the voltage and current of a power line to calculate electrical impedance, tripping a circuit breaker if the impedance drops below a set limit that indicates a fault



The relay continuously monitors line voltage through a potential transformer (PT) and line current through a current transformer (CT). It divides the voltage by the current (Z = V/I) to find the apparent impedance

When a short-circuit fault happens, the current surges and the voltage drops. This sharp drop in the V/I ratio tells the relay that a fault has occurred within a specific distance from the relay point.



Three-Zone Distance Strategy

1

Zone 3 — Backup

Covers adjacent lines with a longer time delay. Acts as remote backup if the primary relay or breaker fails to clear.

2

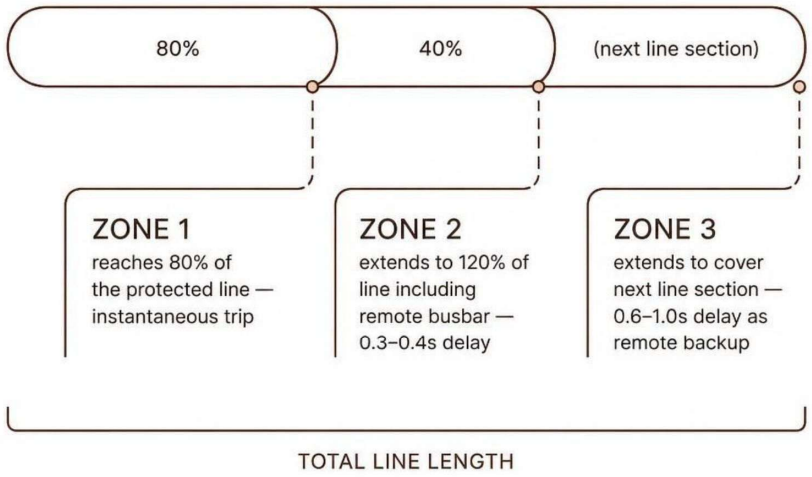
Zone 2 — Extended Reach (120%)

Time-delayed coverage extending to 120% of the protected line length, capturing faults near the remote bus that Zone 1 cannot reach.

3

Zone 1 — Instantaneous (80%)

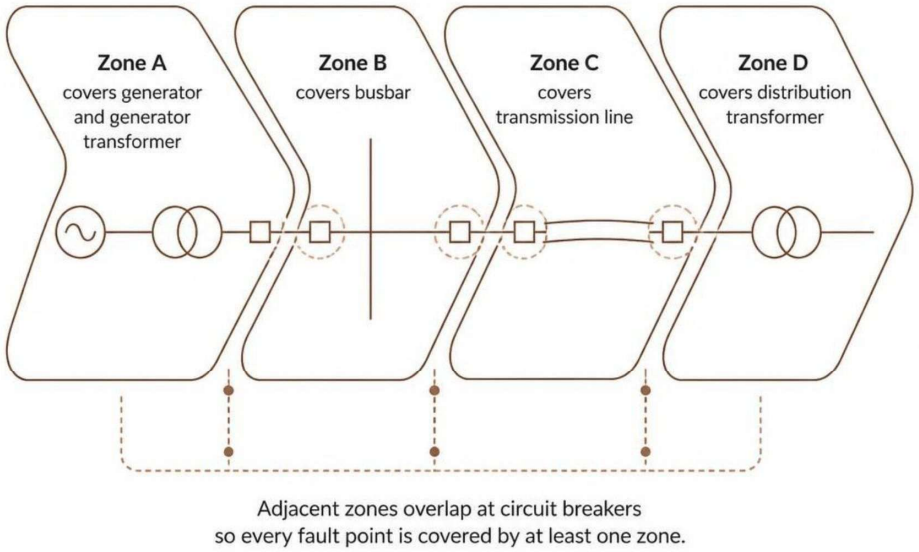
Trips immediately for faults within 80% of the line. The 20% margin prevents overreach caused by measurement errors.



How Distance Relays Work

A distance relay (ANSI 21) continuously calculates **apparent impedance Z = V/I**. Because impedance is proportional to line length, the measured value pinpoints how far along the line a fault lies.

- **Zone 1** — 80% of line, instantaneous trip, no delay
- **Zone 2** — 120% of line, ~0.35 s delay, covers remote bus
- **Zone 3** — Beyond next section, ~0.8 s, back-up role



Why Zones?

Dividing the system into discrete protection zones ensures every fault point is covered by at least one primary relay. **Zones deliberately overlap** at the boundary circuit breakers — so a fault at a zone boundary is seen and cleared by both adjacent zones, eliminating any blind spots.

Challenges in Distance Protection

Fault Type Variation

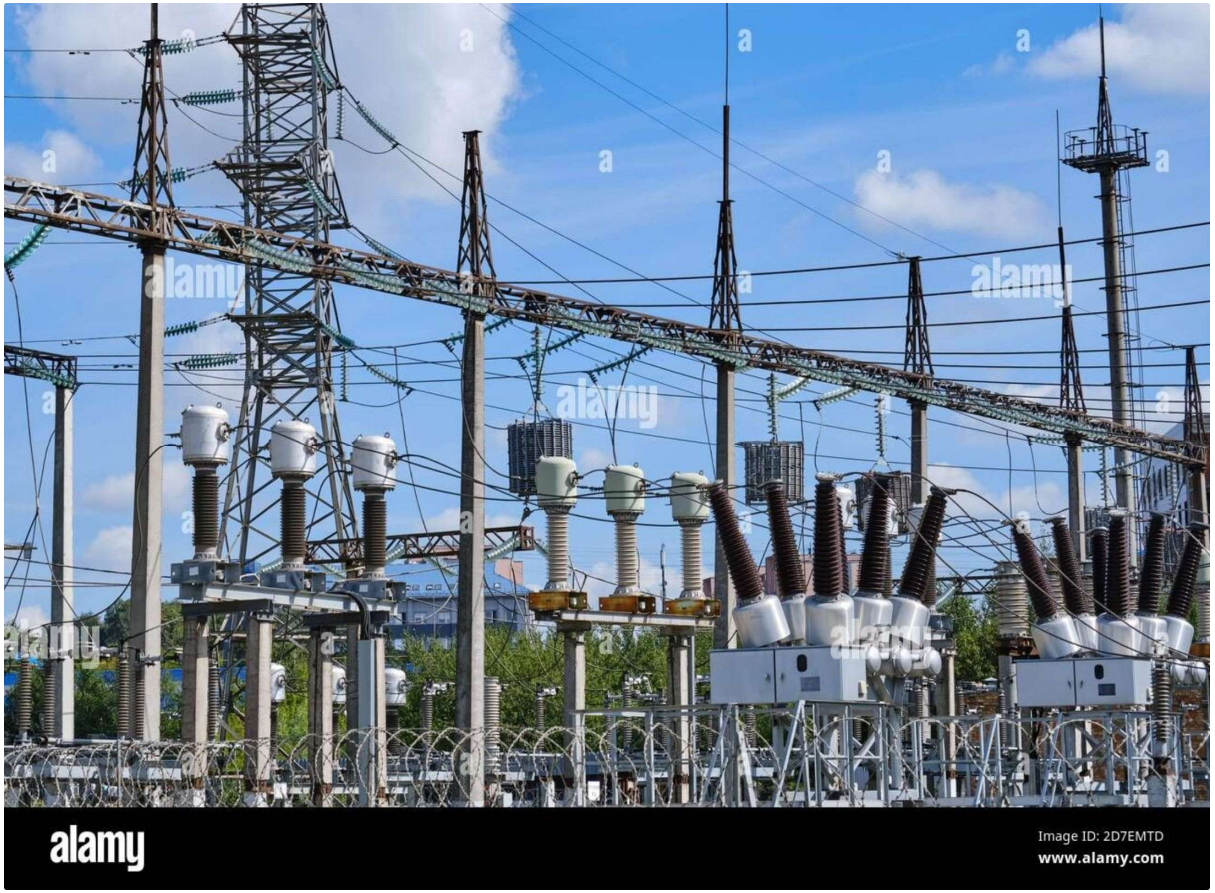
Different fault types (phase-to-phase vs. three-phase) present significantly different apparent impedances, requiring separate relay elements for each.

Line Data Accuracy

Relay reach settings depend entirely on the precision of line impedance data. Errors in data directly translate to errors in reach.

CT/VT Accuracy

Instrument transformer measurement errors can cause under- or over-reach, leading to incorrect operation or failure to trip.





Differential Protection

The most selective and reliable form of protection — designed to guard a specific piece of equipment as a self-contained unit.

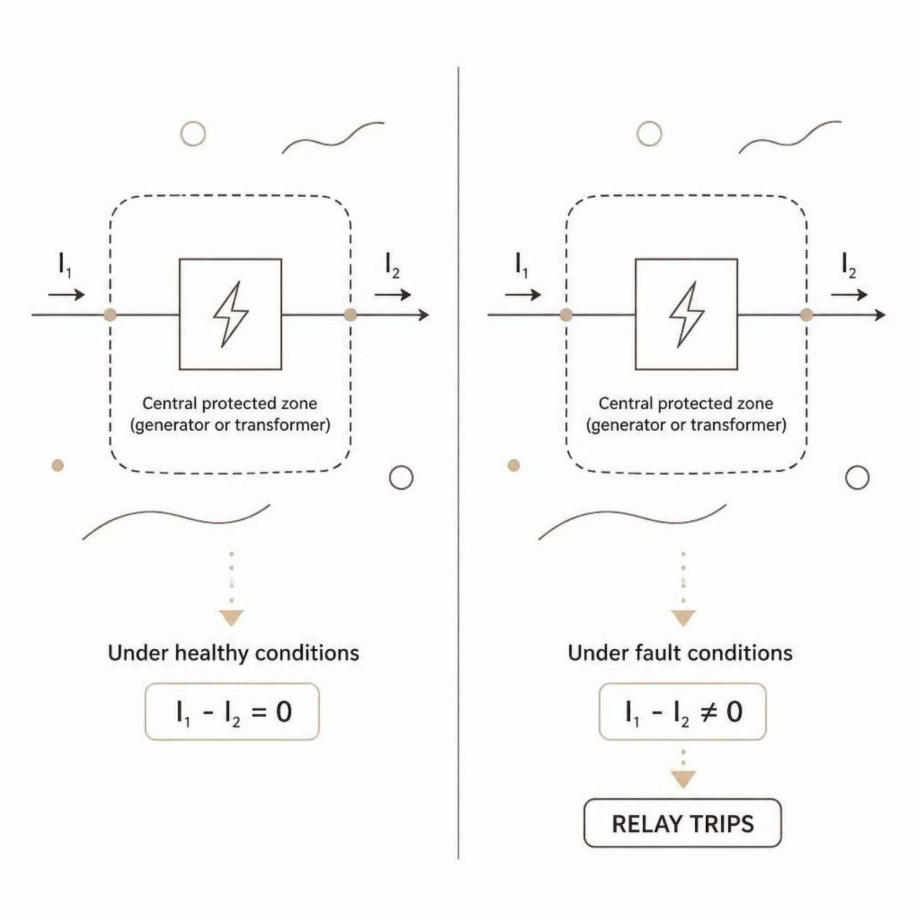
Unit Protection

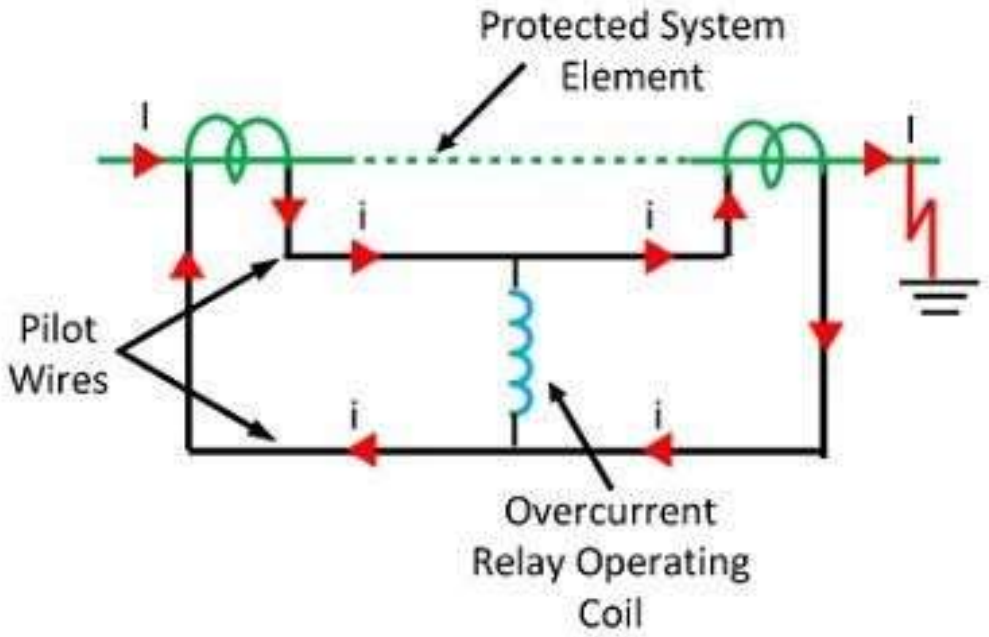
The Kirchhoff Principle

Differential protection is founded on a simple but powerful law: **under healthy conditions, the sum of all currents entering a protected zone must equal zero**. Any imbalance indicates an internal fault, and the relay trips immediately — without needing to coordinate with other relays on the network.

What It Protects

Generators, transformers, busbars, and large motors — any component where the terminals can be individually monitored by current transformers.





For an External Load or Fault

Circuit Globe

Dealing with Errors: The Biased Differential Relay

The CT Mismatch Problem

Simple differential relays can produce a false "spill" current due to unequal CT performance during large external faults — causing unwanted trips.

The Percentage Differential Solution

The biased (percentage) differential relay requires the spill current to exceed a **fixed percentage of the through-current** before tripping. This stabilises the relay during external faults whilst remaining sensitive to genuine internal faults.

Building a Robust Future

The integration of digital, communication-assisted protection schemes ensures **99.99% system availability** on modern high-voltage networks.

Overcurrent

Simple, fast, reliable — the first line of defence on radial systems

Directional

Adds selectivity to networked systems by sensing fault direction

Distance

Impedance-based reach enables precise, coordinated transmission protection

Differential

Gold-standard unit protection — instantaneous, highly selective, immune to network changes

Critical takeaway: Layered protection combining all four principles is the ultimate safeguard against cascading grid failure.



The 50/51 Core Functions

These two ANSI device numbers are the thread running through all relay technologies — from electromechanical to numerical.

ANSI 50 — Instantaneous Overcurrent

Operates with **no intentional time delay**. Trips immediately when current exceeds the set threshold. Used for high-magnitude close-in faults where speed is paramount.

ANSI 51 — Time Overcurrent

Applies an **intentional inverse time delay** via a time-current characteristic (TCC) curve. Higher currents produce shorter operating times, enabling graded coordination across multiple protection zones.



Electromechanical Relays

From the early 1900s through to the 1990s, electromechanical relays were the industry standard. These devices use induction discs, coils, mechanical contacts, and solenoid elements to produce protective timing and trip behaviour — entirely without electronics.

Induction-Type Relay: Key Components

Two Elements, One Relay

A traditional induction-type overcurrent relay combines two distinct operating elements:

For complete three-phase protection, a minimum of **three discrete relay units** were required — one per phase.

Induction Disc (51)

Time-overcurrent element. Disc rotation speed is proportional to current magnitude, providing an inverse-time characteristic.

Solenoid / Attracted Armature (50)

Instantaneous element. Operates immediately when current exceeds pickup, with no disc involved.

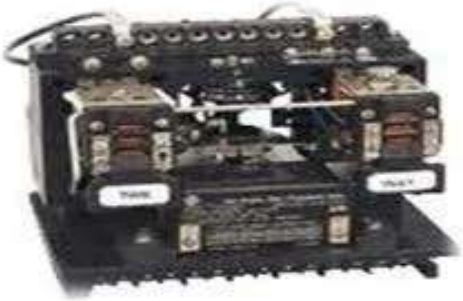
Electromagnetic Relay

Electromagnetic relays are those relay which operates on the principle of electromagnetic attraction.

It is a type of magnetic switch that uses a magnet to create a magnetic field.

The magnetic field is then used for opening and closing the switch and for performing the mechanical operation







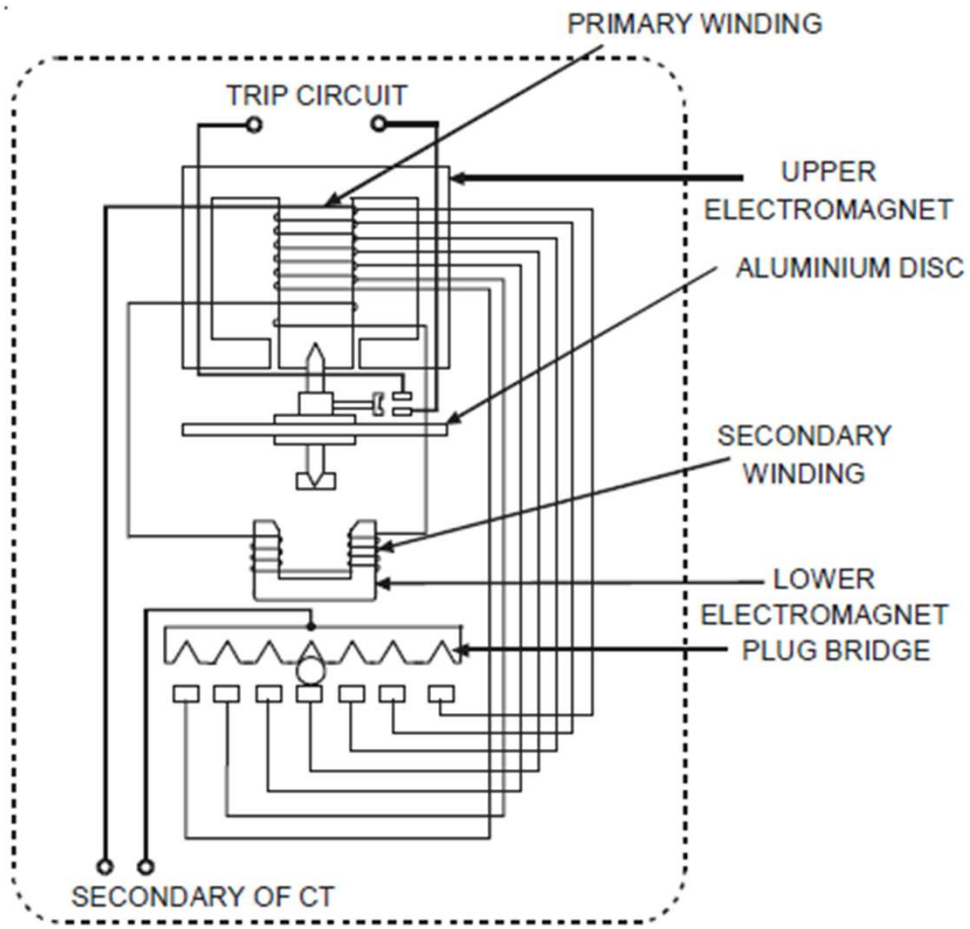


Electromagnetic Induction Relay

The electromagnetic relay operates on the principle of a split-phase induction motor.

The initial force is developed on the moving element, which may be a disc or another form of the rotor of the non-magnetic moving element.

The force is developed by the interaction of electromagnetic fluxes with the eddy currents induced in the rotor by these fluxes.


PRIMARY WINDING
TRIP CIRCUIT
UPPER
ELECTROMAGNET
ALUMINIUM DISC
SECONDARY
WINDING
LOWER
ELECTROMAGNET
PLUG BRIDGE
SECONDARY OF CT

Induction-Type Overcurrent Relay

- The relay consists of two electromagnets: an upper electromagnet and a lower electromagnet.
- The upper electromagnet has two windings:
  - The primary winding is connected to the secondary of a CT on the protected line.
  - The primary winding is tapped at intervals to provide current-setting adjustment.
  - An EMF is induced in the second winding by transformer action from the primary winding.
- The second winding of the upper electromagnet is connected in series with the winding of the lower electromagnet. The induced EMF causes current to flow through this series circuit.
- The upper and lower electromagnets produce leakage fluxes displaced in space and phase, creating rotational torque on the aluminium disc.
- The relay torque is:

$$T = K_1 I_{\text{rms}}^2 - K_2$$

where I_{rms} is the relay-coil current and K_2 represents the restraining torque of the spring.

- When the operating torque exceeds the restraining torque, the disc rotates through a preset angle.
- The moving contact on the disc spindle bridges the fixed contacts, completing the trip circuit and initiating circuit-breaker operation.

Different types of structures have been used for obtaining the phase difference in the fluxes. These structures are

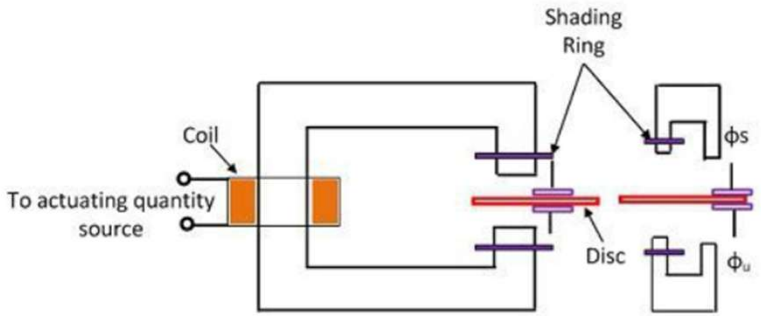
a. Shaded pole structure
b. Watt-hour meter or double winding structure
c. Induction cup structure.

a. Shaded pole structure

-**Coil Energization:**
- Utilizes current flowing through a single coil.
- Wound on a magnetic structure featuring an air gap.

Air-gap Fluxes:
- Produced by the initializing current.
- Split into two fluxes that displace in time-space due to a shaded ring.

Shaded Ring:
- Composed of a copper ring encircling part of each pole face.



Shaded Pole Construction of Induction Disc Relay

Aluminum Disc Characteristics:
- Made of aluminum with low inertia.
- Requires less deflecting torque for movement.

Induction Mechanism:
- Two rings have current induced by the alternating electromagnetic flux.
- Magnetic field from the current causes flux in the iron ring to lag phase by 40° to 50° behind the flux in the unshaded portion of the pole.

Watt-Hour Meter Structure

- Consists of an E-shaped electromagnet, a U-shaped electromagnet, and a rotating disc.
- The disc is free to rotate in the air gap between the two electromagnets.
- Different resistance and inductance values in the magnet circuits produce the required phase displacement.

E-shaped electromagnet:

- Contains primary and secondary windings.
- Primary winding carries the relay current I_1.
- Produces magnetic flux ϕ_1.

Secondary winding:

- Connected to the U-shaped electromagnet.
- Induced EMF in the secondary causes current I_2 to flow.
- The U-shaped electromagnet produces flux ϕ_2.

Watt-Hour Meter – Operating Principle

The two magnetic fluxes are displaced in phase by an angle θ:

$$\phi_1 \xleftrightarrow{\theta} \phi_2$$

Their interaction produces driving torque on the disc:

$$\boxed{T \propto \phi_1 \phi_2 \sin\theta}$$

Thus, torque requires:

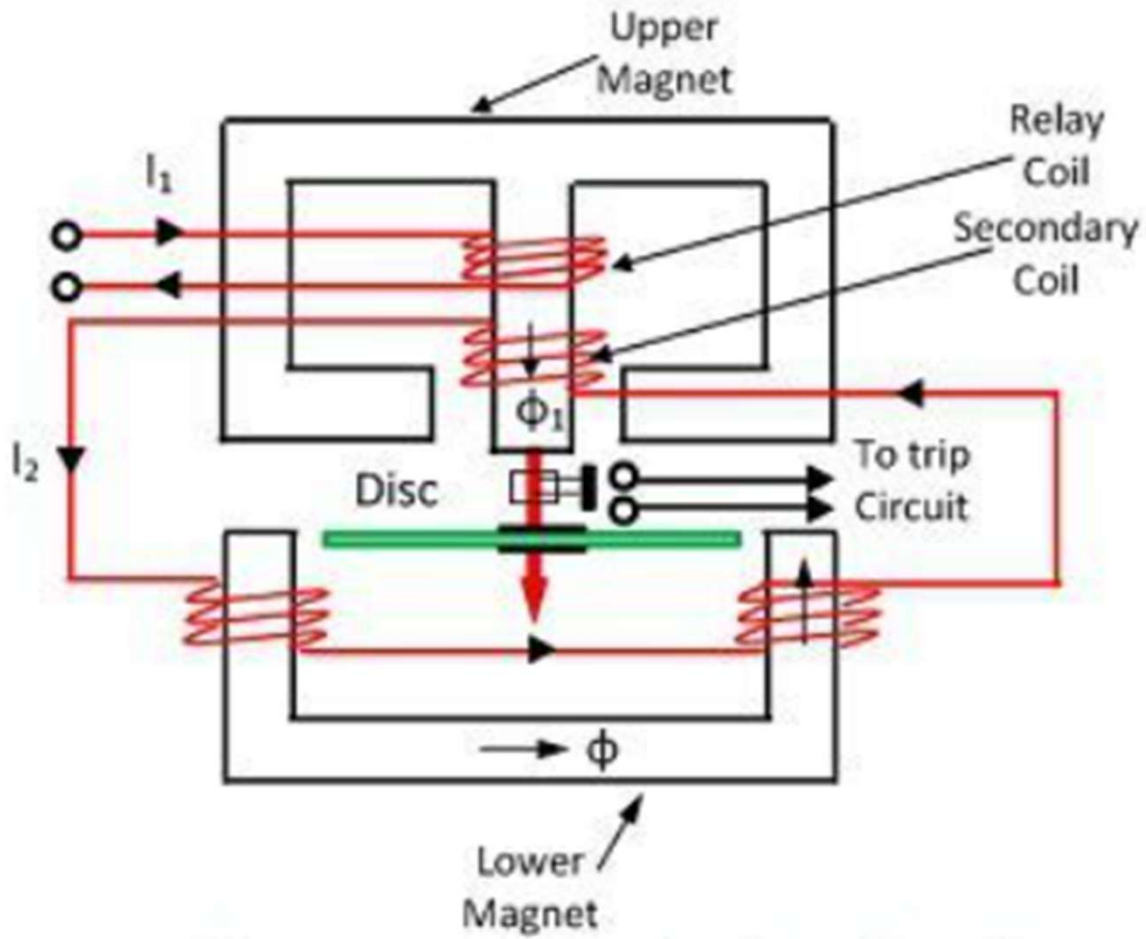
- Two alternating magnetic fluxes
- A phase difference between the fluxes
- A conducting disc in the magnetic field

Relay operation can be controlled through the secondary circuit:

$$\boxed{\text{Secondary Closed} \rightarrow I_2 \rightarrow \phi_2 \rightarrow \text{Torque}}$$

$$\boxed{\text{Secondary Open} \rightarrow I_2 = 0 \rightarrow \phi_2 = 0 \rightarrow T = 0}$$

Hence, opening the secondary circuit prevents torque from developing and makes the relay inoperative.



Watt-hour Meter Type Induction Disc Relay

Induction Cup Relay

- An induction cup relay operates on the principle of electromagnetic induction.
- It consists of two or more electromagnets energized by the relay coils.
- A stationary iron core is placed between the electromagnets.
- A light conducting cup, usually made of aluminium, is free to rotate around the stationary core.

Operating principle:

- The electromagnets produce alternating magnetic fluxes displaced in phase.
- These fluxes produce a rotating magnetic field.
- The rotating magnetic field induces eddy currents in the conducting cup.
- Interaction between the magnetic field and induced currents produces electromagnetic torque.
- The resulting torque causes the cup to rotate and operate the relay contacts.

$$\boxed{\text{Rotating Magnetic Field} \rightarrow \text{Induced Current} \rightarrow \text{Torque} \rightarrow \text{Cup Rotation}}$$

Induction Cup Relay – Features

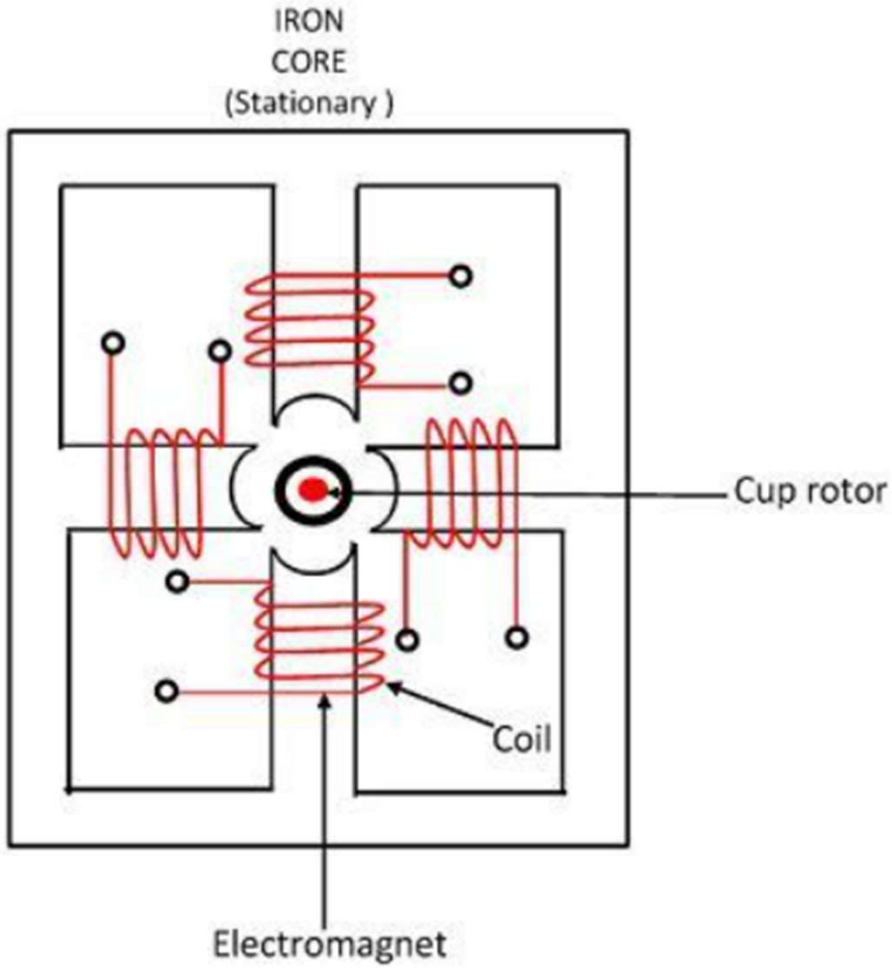
- Produces higher torque than shaded-pole and watt-hour meter type induction relays.
- The rotating cup has very low inertia, resulting in fast operation.
- Typical operating time can be approximately:

$$\boxed{t_{\text{op}} \approx 0.01 \text{ s}}$$

- High torque-to-inertia ratio makes it suitable for high-speed protective relaying.
- Commonly used in directional, distance and other induction-type protective relays.

Key advantage:

$$\boxed{\text{High Torque} + \text{Low Inertia} \rightarrow \text{Fast Relay Operation}}$$

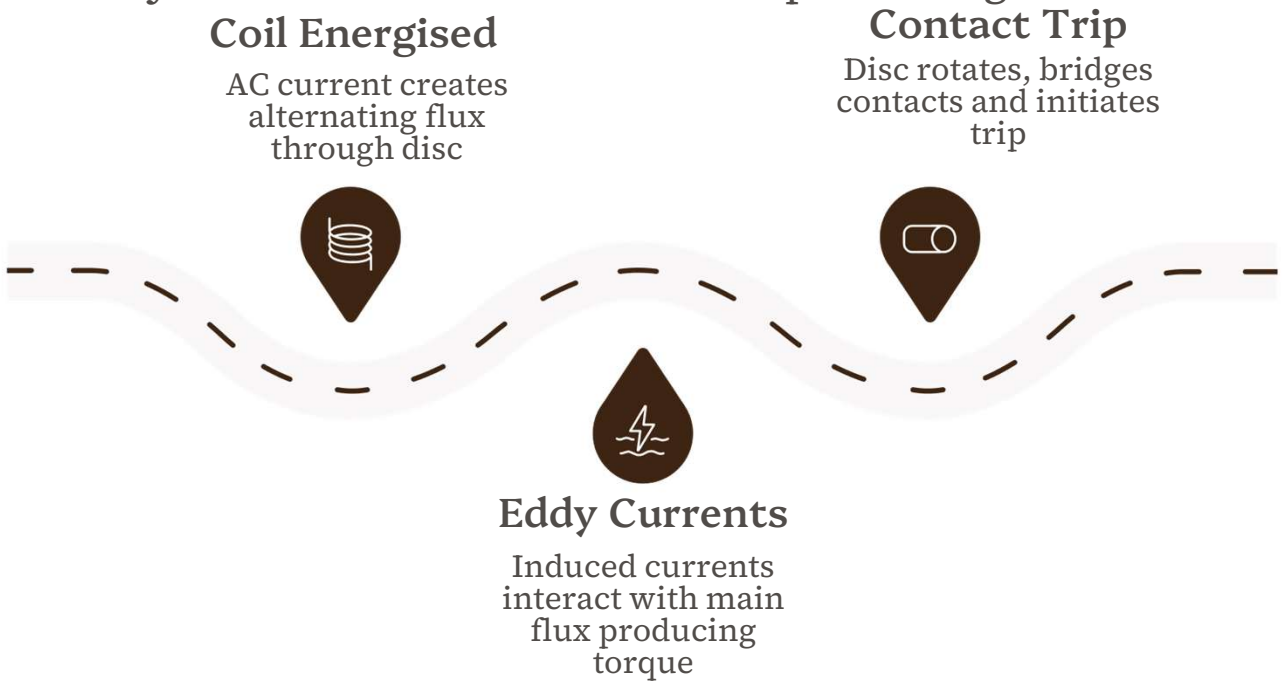


Induction Cup Relay

Circuit Globe

Induction Disc Physics

The operating principle mirrors that of an induction motor —
Torque is produced by the interaction of two out-of-phase magnetic fluxes.



The disc rotates at a speed proportional to the square of the current magnitude.
A **permanent magnet** provides braking torque to ensure smooth, stable rotation and prevent overshoot.

Key Settings: Electromechanical 51

1

Current Tap (Pickup)

Selects the CT secondary current at which the relay begins to pick up. The tap determines field strength in the coil and sets the minimum operating current. Typical taps: 0.5 A to 12 A on 5 A CT secondary.

2

Time Dial

Adjusts the disc travel distance, directly shifting the time-current curve up or down. Higher TD = slower operation at any given multiple of pickup current.

3

Curve Shape

Fixed by the relay model (e.g., Standard Inverse, Very Inverse, Extremely Inverse). Unlike numerical relays, the electromechanical curve shape **cannot be changed in the field** — a different relay model is required.

Relay Curves & Coordination Time

A time-current characteristic curve shows **relay operating time only** — it does not include the full fault-clearing sequence. Total fault-clearing time must account for all delays in the protection chain.

1

CT Accuracy Margin

~12 cycles safety factor to account for CT ratio errors and transient saturation effects.

2

Relay Operating Time

Time read directly from the TCC curve at the multiple of pickup current experienced during fault.

3

Breaker Trip & Clear

Typically 3–5 cycles for a modern circuit breaker to interrupt current after receiving the trip signal.

4

Interposing Delays

Auxiliary relay and wiring delays add a further 1–3 cycles to the total fault-clearing time.

Coordination Interval: The 0.4 s Rule

The **minimum coordination interval** for electromechanical relays is typically **0.4 seconds (24 cycles)**. This interval must be maintained between the operating curves of any two adjacent relays to guarantee selectivity — ensuring the downstream relay operates before the upstream relay can close its contacts.

This 0.4 s margin is a direct consequence of the mechanical tolerances, CT errors, and breaker clearing times inherent to electromechanical protection systems.

Static Relays

Static relays emerged in the 1960s–70s, replacing moving mechanical parts with **analogue electronic circuits** — transistors, operational amplifiers, and RC timing networks. They replicate the time-current behaviour of induction disc relays without any rotating disc, offering improved repeatability, reduced maintenance, and faster reset times.



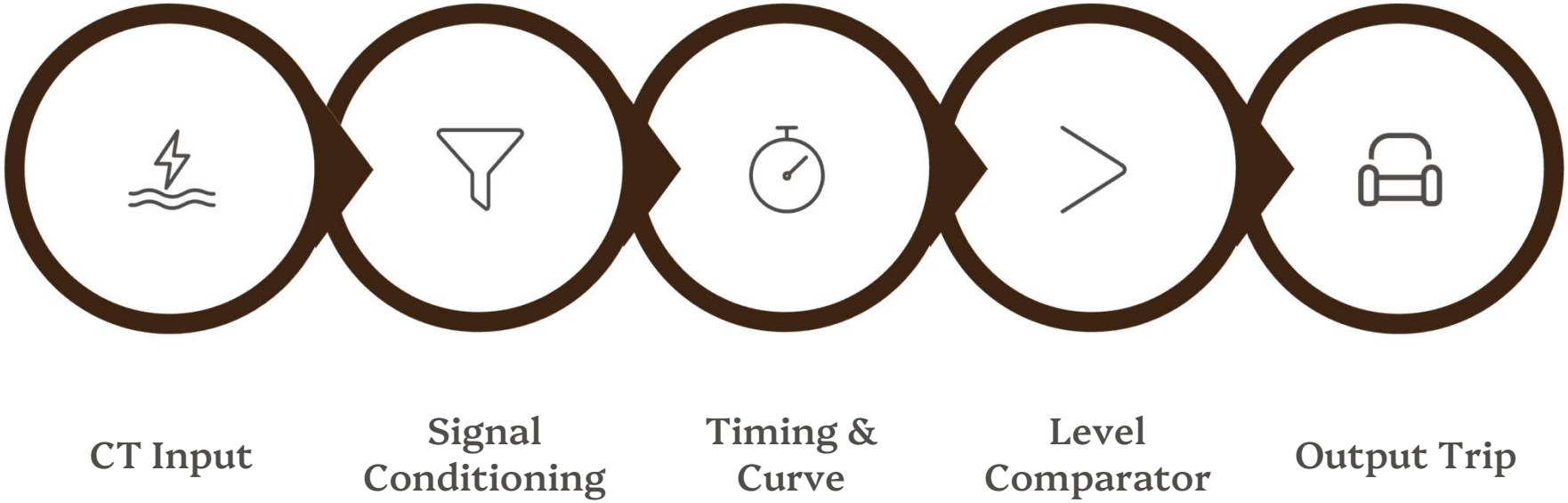

6 5 4 3 2 1
PCS-700
POWER CONTROL SYSTEM
NO C NC
SUPPLY
FAULT
50 HZ.
55 HZ.
OVER FREQUENCY RELAY
230VAC
NO C NC


OVER CURRENT RELAY
OCSPL
5A
HEALTHY
FAULT
TEST
RESET
Pyok dv's
BANGALORE

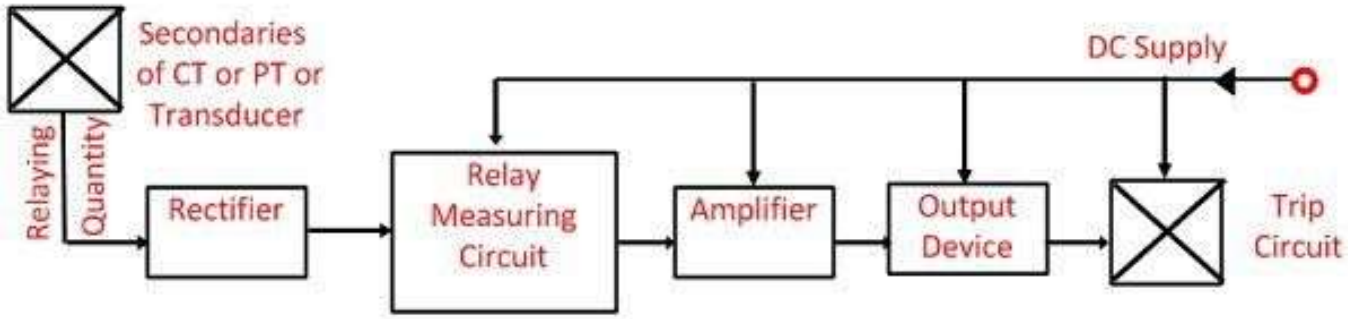

SOLID STATE RELAY

Static TOC (51): Block Diagram

The signal processing chain of a static time-overcurrent relay follows a consistent architecture across manufacturers.



Each stage is implemented with discrete analogue components — no software, no sampling, no microprocessor involved.



Block Diagram of Static Relay

Circuit Globe

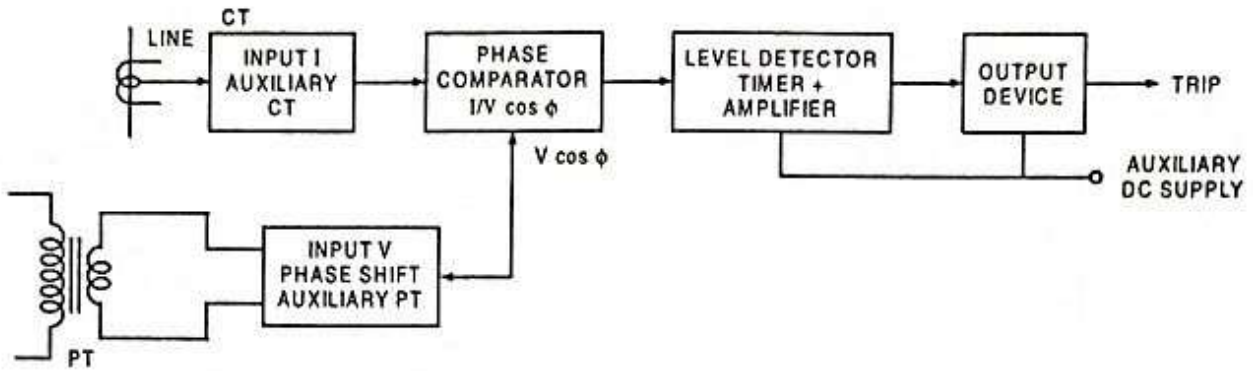
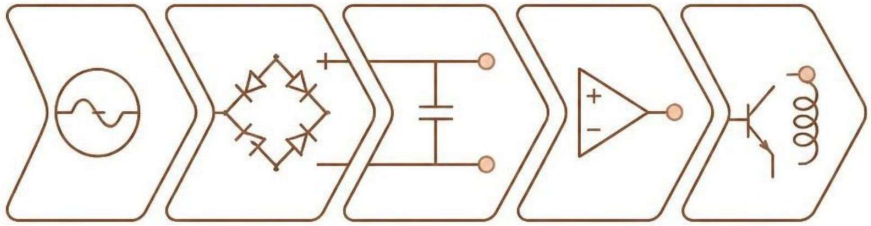


Fig. 10.39. ***Block Diagram of Directional Static Overcurrent Relay***

Static Instantaneous (50): Block Diagram

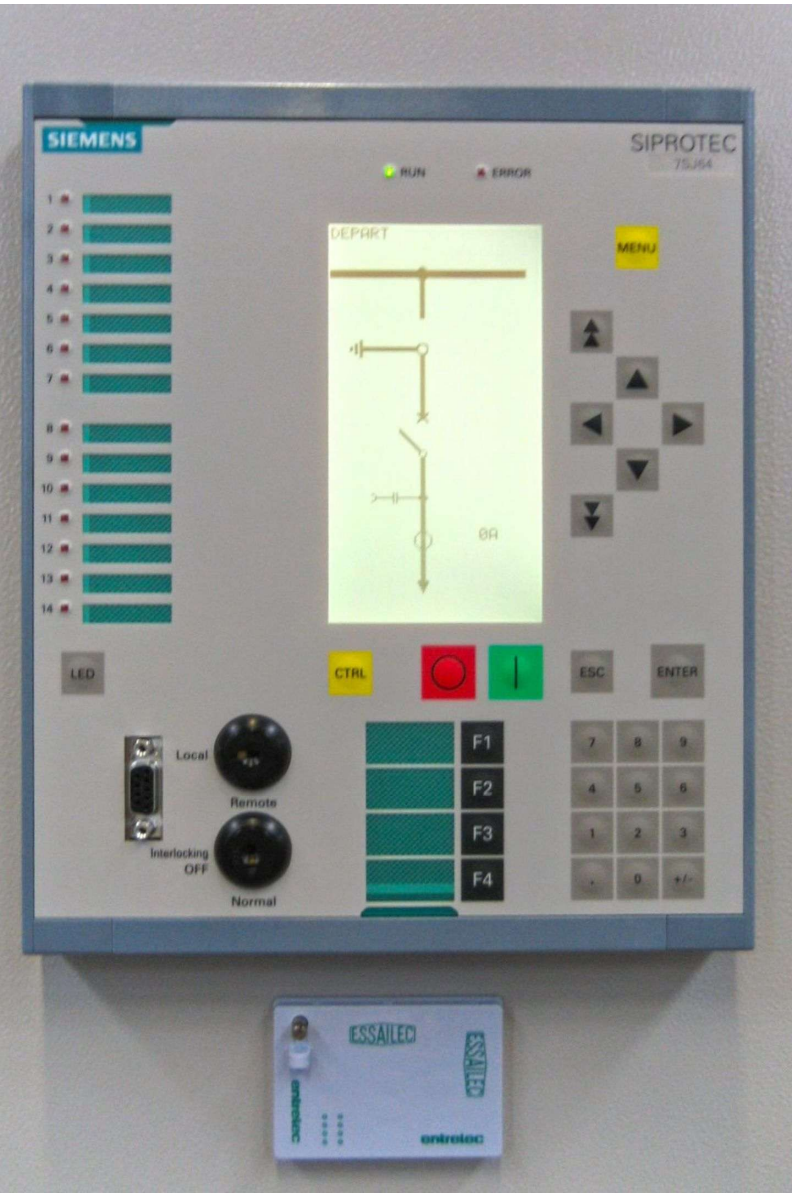


Instantaneous Element Logic

The static 50 element is considerably simpler than the 51 — it contains no timing network. The key signal path is:

Current → Rectifier → Level Comparator → Instantaneous Trip Output

- No RC timing circuit — comparison is direct and immediate
- Pickup threshold set by an adjustable reference voltage
- Operating time is typically **8–16 ms** (half to one cycle)
- Output transistor drives an interposing output relay to trip the breaker



Microprocessor & Numerical Era

From the 1980s onwards, microprocessor-based and fully numerical relays replaced analogue circuits with **sampled digital signals and software algorithms**. A single numerical relay can implement dozens of protection functions simultaneously — overcurrent, distance, differential, and more — in one compact, communicating device.


Microelettrica Scientifica
MOTION
STALL
START
INHIBIT
MULTIFUNCTION
MOTOR
PROTECTION
RELAY TYPE
MM30-743
MODE
SELECT
+
-
PROG.
ENTER/RESET

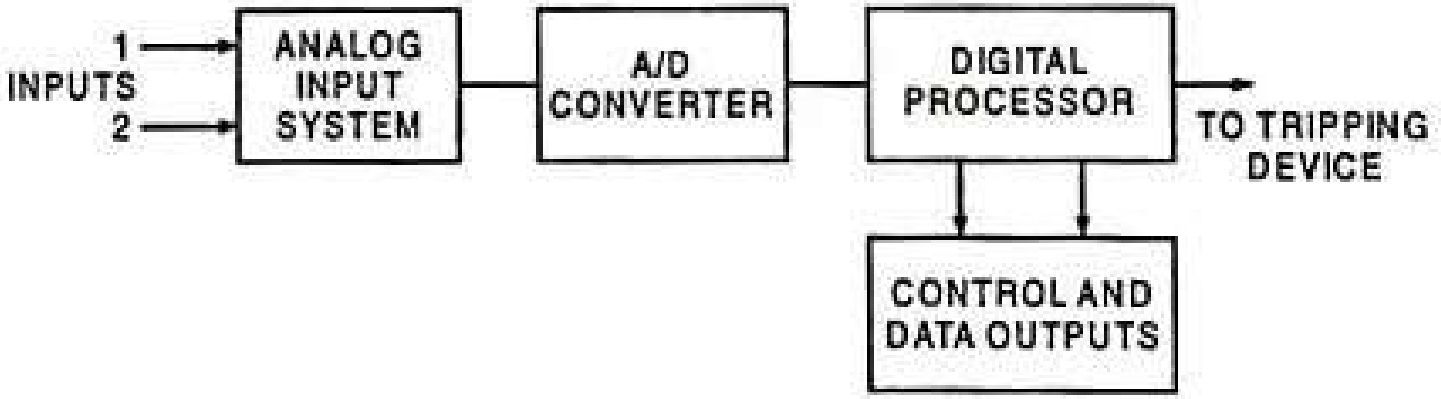
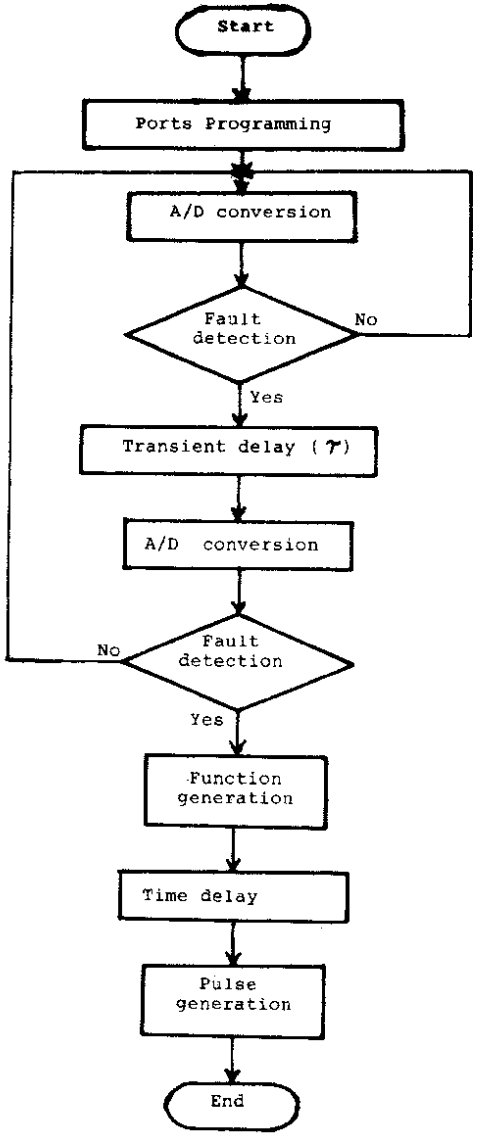
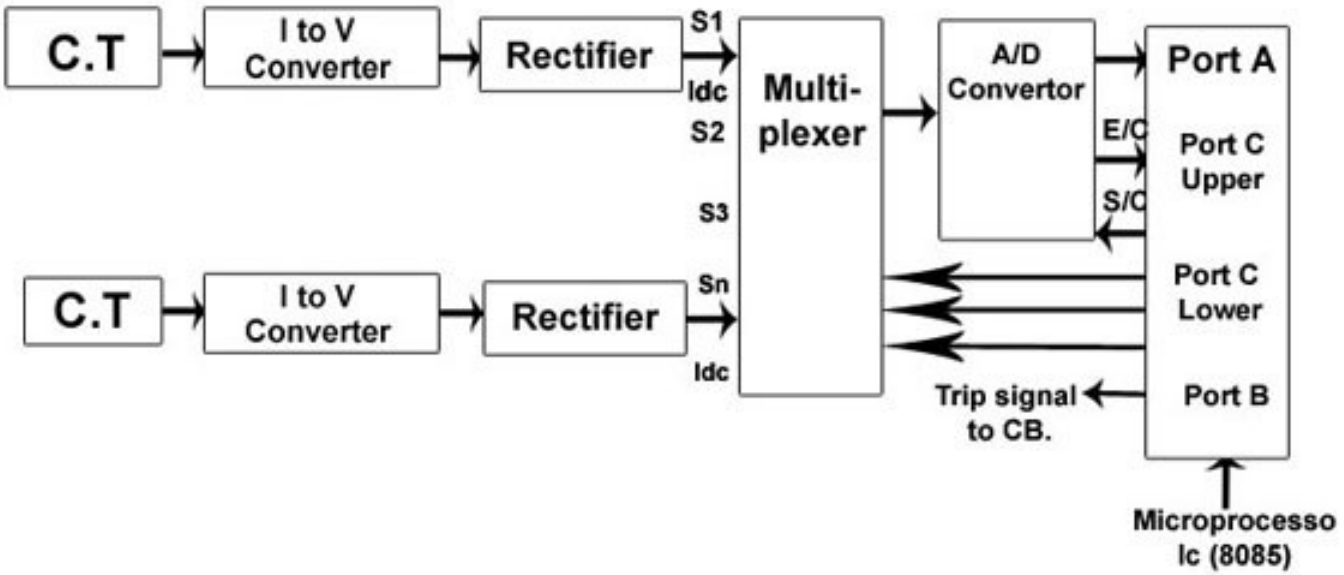


Fig. 10.58. *Basic Protection Scheme Using Microcomputer*

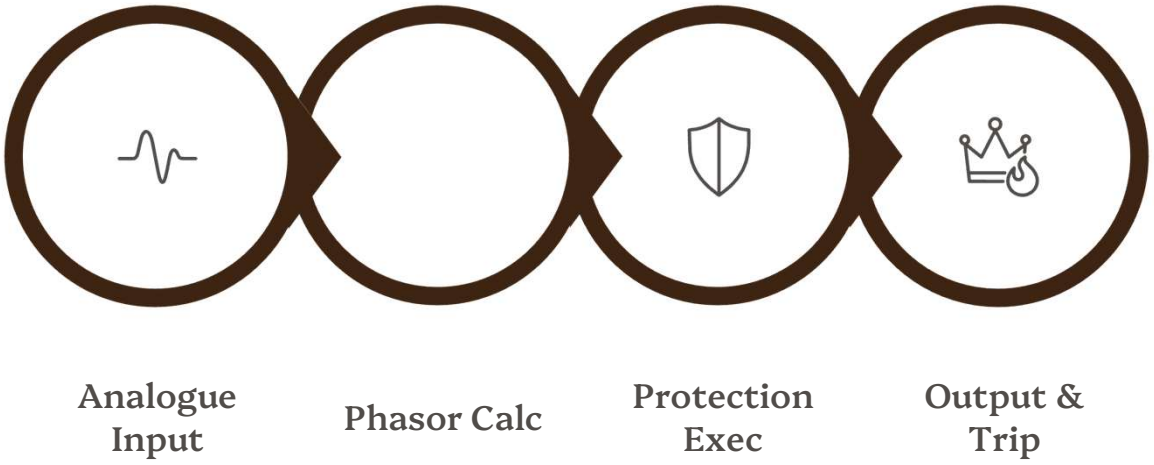


Flow chart of the software in the microprocessor-based overcu relays.



Block diagram of a microprocessor-based overcurrent relay

Microprocessor-Based 50/51: Building Blocks



Key Processing Steps

- **Sample & Convert**

  Current waveform sampled at 16–64 samples/cycle and converted to digital values by the ADC.

- **Phasor Extraction**

  DFT algorithm computes fundamental frequency magnitude, filtering harmonics and noise.

- **50/51 Decision**

  Magnitude compared against pickup threshold; 51 timing accumulated per the curve equation each cycle.

- **Output & Record**

  Trip output energised; event log, fault record, and waveform capture stored for post-fault analysis.

Numerical 51: Curve Parameters & Settings

Numerical relays implement IEC 60255 and IEEE C37.112 standard curve equations directly in software, offering far greater flexibility than electromechanical models.

Pickup (Tap)

Set as a multiple of CT secondary current. Current must exceed pickup for the timing element to begin accumulating. Fine resolution (e.g., 0.01 A steps) vs. fixed electromechanical taps.

Time Dial (TDS / TMS)

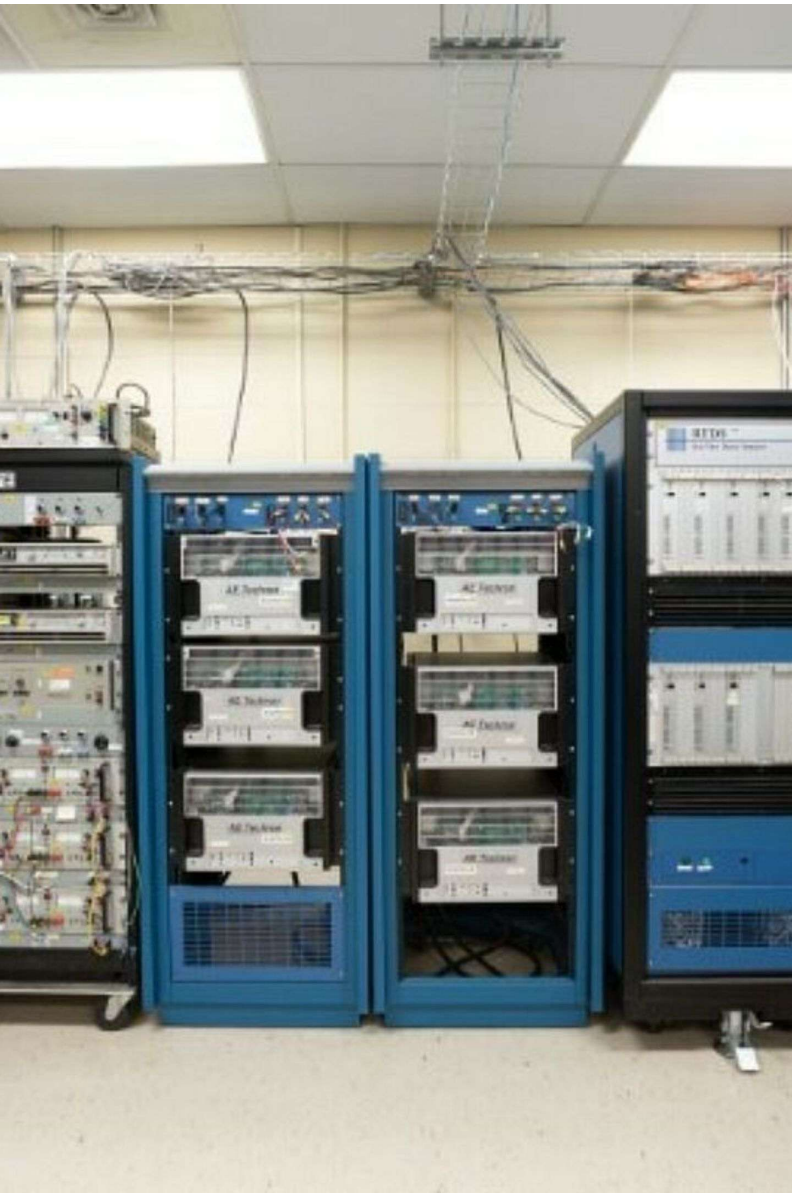
Multiplies the curve equation output. Increasing TDS moves the entire curve **upward** — increasing operating time at every current level. Numerical relays allow small increments (e.g., 0.01 steps vs. discrete positions).

Curve Shape

Selectable from a list: Standard Inverse, Very Inverse, Extremely Inverse (IEC), Moderately / Very / Extremely Inverse (IEEE). Any curve is available in one device — no hardware change required.

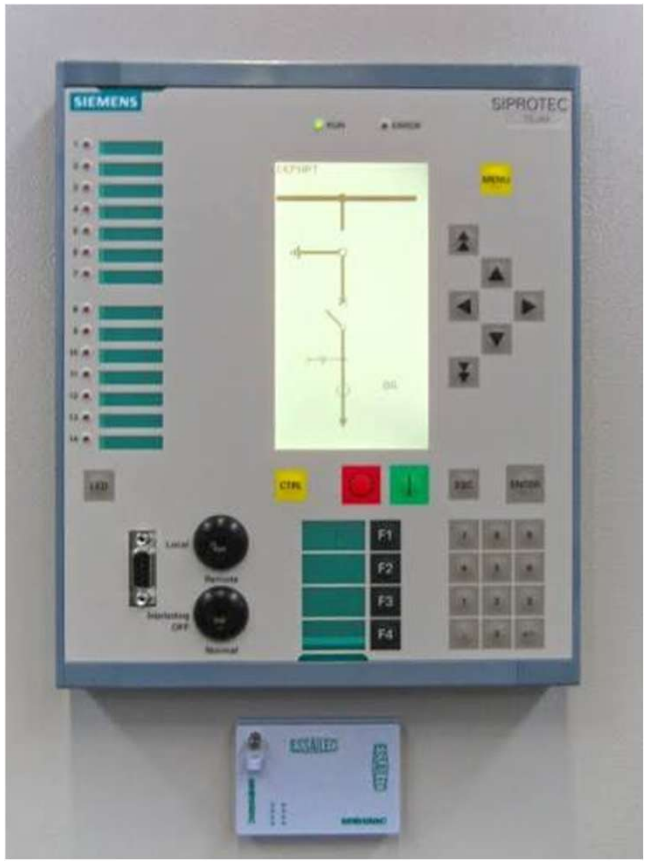
Reset Mode

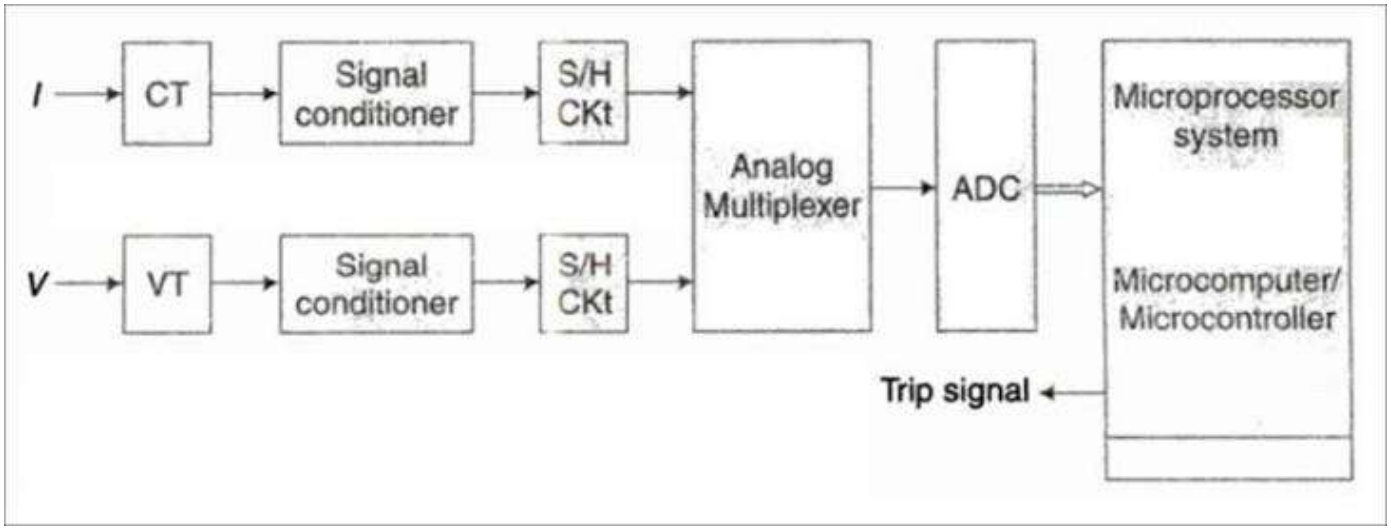
Electromechanical-style timed reset or instantaneous (electronic) reset — selectable to match coordination with upstream electromechanical relays during retrofit projects.

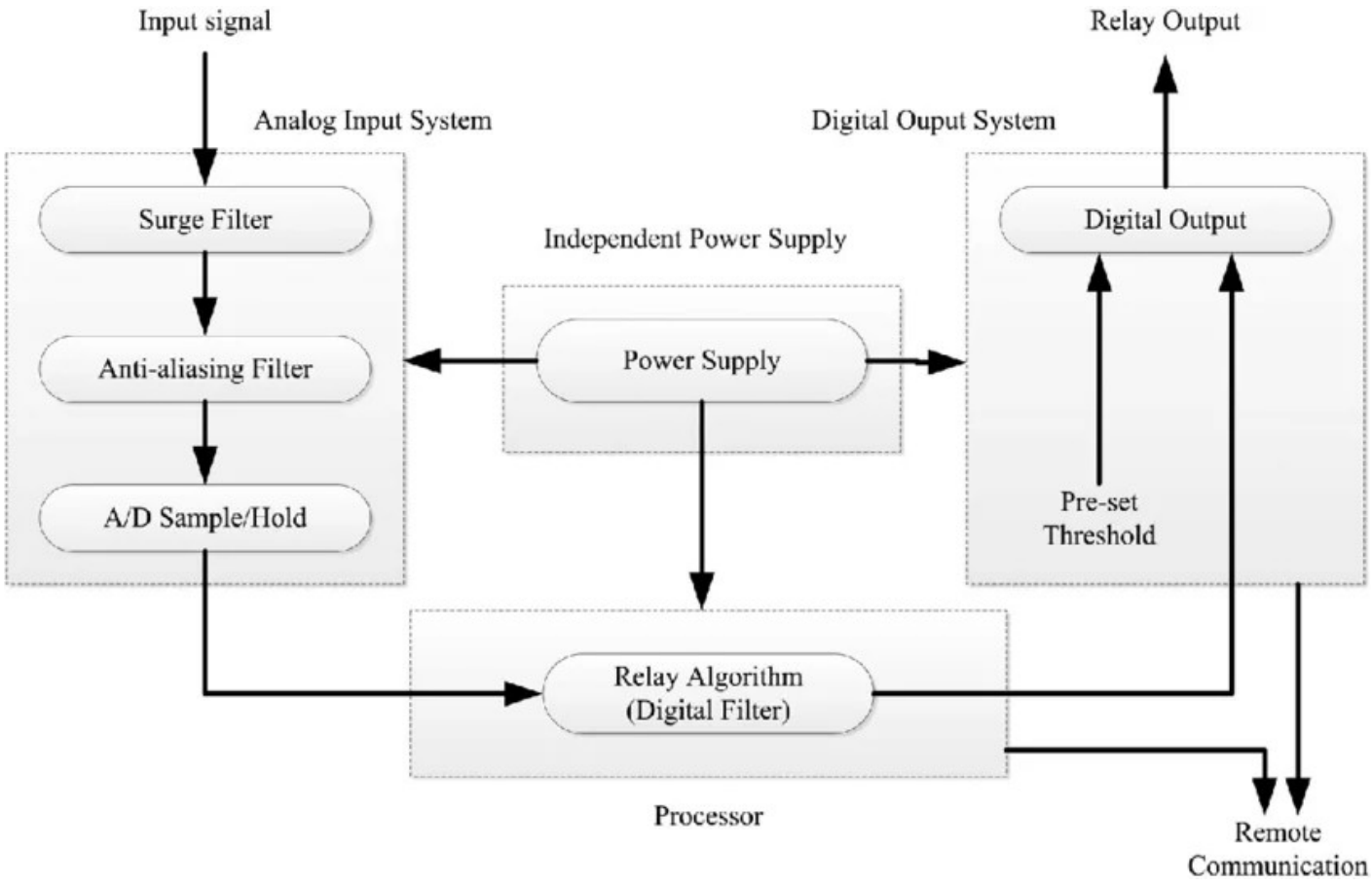


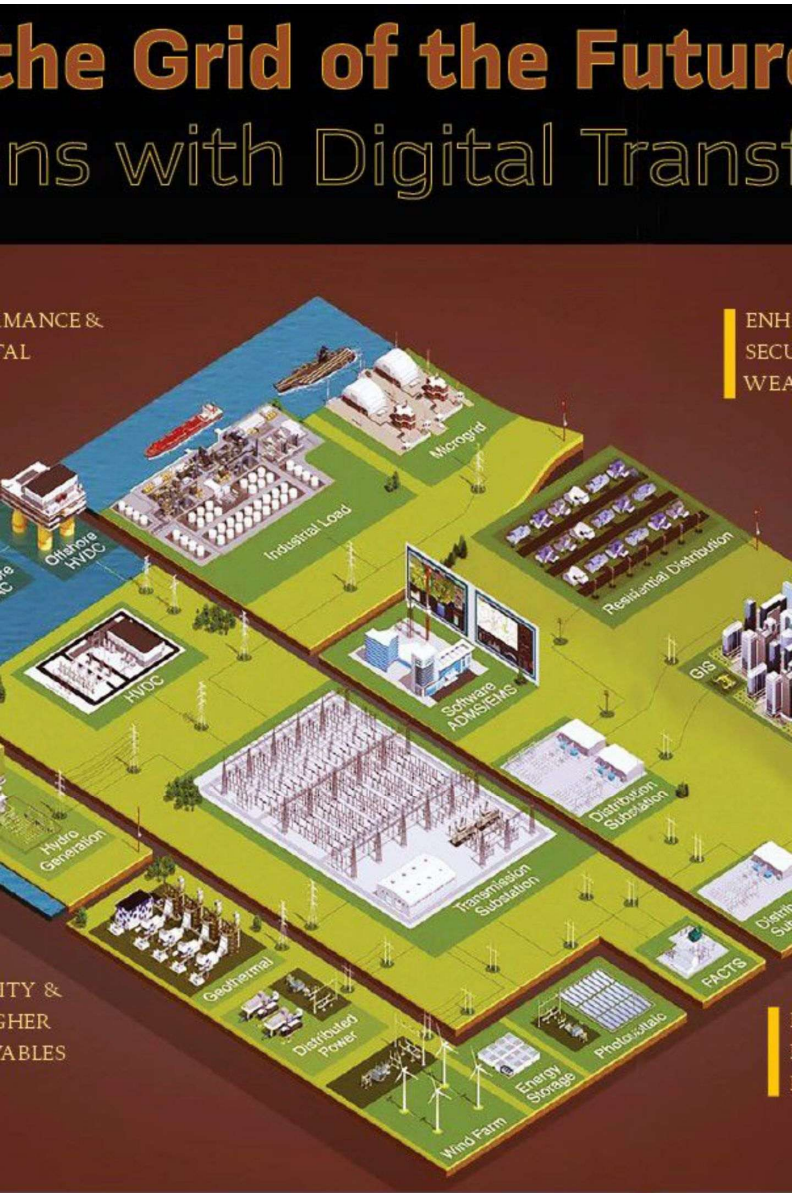
Numerical Relays & Beyond

Numerical relays represent the transition from hardwired analogue logic to fully software-defined protection — replacing discrete components with microprocessors, DSP chips, and configurable algorithms. The result is a single device capable of delivering **multi-functional protection, monitoring, metering, and control** within one standard 4U panel chassis.


SIEMENS
SIPROTEC
RUN
ERROR
MENU
LED
CTRL
ESC
ENTER
Local
Remote
Interlocking OFF
Normal
F1
F2
F3
F4


I
CT
Signal conditioner
S/H CKt
V
VT
Signal conditioner
S/H CKt
Analog Multiplexer
ADC
Microprocessor system
Microcomputer/ Microcontroller
Trip signal


Input signal
Analog Input System
Surge Filter
Anti-aliasing Filter
A/D Sample/Hold
Independent Power Supply
Power Supply
Relay Output
Digital Ouput System
Digital Output
Pre-set
Threshold
Relay Algorithm
(Digital Filter)
Processor
Remote
Communication



The Future of Grid Protection

The protective relay has evolved from a mechanical disc into an intelligent grid sensor. The next evolution is already underway.

1 — **Electromechanical**

Induction disc, fixed curves, 0.4 s coordination intervals, one function per device.

2 — **Static & Microprocessor**

Analogue circuits and sampled digital signals. Multiple functions, improved repeatability, event recording.

3 — **Numerical**

Software-defined protection, IEC 61850, waveform capture, self-diagnostics, and remote access.

4 — **Adaptive & AI-Driven**

Real-time adaptive settings, predictive fault analysis, machine learning, and the relay as an active smart grid data node.